

Are Designated Market Makers Necessary in Centralized Limit Order Markets?¹

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Abstract

This paper models price formation and order placement strategies in a dynamic centralized limit order market with designated non monopolist market makers. This market design differs from the NYSE and has been adopted by several stock exchanges in Europe and North America mainly for the trading of mid caps. The model is constructed as an extension of Foucault (1999) in order to include market makers who are required to provide two-sided quotes in the book. Traders can place either a market order or a limit order and, in the latter case, they can choose the order's limit price. Comparison between a limit order market with market makers and a pure limit order market yields several testable implications. First, the clearing frequency is higher in a limit order market with designated market makers. Second, the best quotes in the limit order book with market makers are either more or less attractive than in a pure order book depending on whether the best quote has been placed by a trader or a market maker. On an ex ante basis, we find that expected best quotes in the order book can be more or less favourable in the presence of market makers depending on the asset volatility and the differences in valuation among traders. The model thus yields empirical predictions relative to the added value of non monopolist market makers across assets and/or periods. Other things equal, expected best prices in a limit order book with market makers become more attractive than in a pure limit order book when the asset volatility increases. Since asset volatility decreases with equity capitalisation, our results provide a rationale for the trading of mid caps being organized as limit order books with designated dealers. By contrast, the presence of market makers on large capitalisation stock markets could deteriorate market liquidity.

Keywords : Market Microstructure, Limit Order Books, Designated Market Makers

1 Introduction

Most stock exchanges, while mainly organized as limit order markets, have market makers as part of their market design. The main purpose of market making is to provide liquidity to the markets. However, the way market makers are integrated to the trading system can differ significantly across exchanges. One market model features a hybrid specialist/limit order market like the NYSE. Because he has privileged information over other traders on future order flow or the limit order book, the NYSE-like specialist is often called the monopolist. A number of stock exchanges have adopted an alternative market design in which non monopolist dealers¹ and traders interact in a centralized limit order market.

As opposed to the NYSE specialist, market makers in these limit order markets do not possess any monopolistic information that the other traders do not have. They are required to maintain a regular presence and guarantee a minimum level of liquidity by posting limit orders to buy and sell throughout the trading day. Examples include the London Stock Exchange SETSmm² trading mechanism for mid caps, the Toronto Stock Exchange³, the Euronext⁴ and Italian Stock Exchange trading systems for small and mid cap stocks, among others. More recently, the Australian Stock Exchange has made proposals in favor of the introduction of non monopolist dealers on small and mid cap stocks limit order markets. These trading mechanisms have mainly being implemented for the trading of small and mid caps rather than large cap stocks. The rationale behind having market makers for small and mid caps only is that these are less liquid securities than large caps and thus those which most need the presence of market makers⁵.

By placing two-sided quotes in the book, non monopolist market makers enter in competition

¹The terms dealers and market makers are used interchangeably in this paper.

²Launched at the end of 2003, this trading mechanism has replaced the SEAQ (Stock Exchange Automation Quotation) dealer market for FTSE 250 securities which are not liquid enough to be traded on a pure limit order book.

³Market makers called Registered Traders are part of the Toronto Stock Exchange limit order market.

⁴Euronext has introduced Liquidity Provider Agreements on small and mid caps in order to allow dealers to provide continuous liquidity in the central order book. The parties in these contracts are the stock exchange and a market maker firm.

⁵Promotional presentation of the LSE's SETSmm trading system is written in the following terms : "It's clear that order driven trading has become the standard for major market users in the most liquid stocks. But there is, just as clearly, still a strong need for dedicated market making in the mid-cap securities where there are different fundamental market dynamics" (Source : www.londonstockexchange.com/setsmm).

with limit order traders⁶ for the execution of limit orders. How such competition affects price formation in limit order markets has recently received an empirical interest (Nimalendran and Petrella (2003), Mann, Venkataraman and Waisburd (2004)). By contrast, while NYSE-based trading systems have been extensively analyzed, to our knowledge, there exists no theoretical paper that explores the link between the presence of designated non monopolist dealers, traders' optimal strategies and price formation in limit order markets. This paper constitutes a first attempt to adress these issues. More precisely, we seek to answer the following questions.

- How the presence of designated dealers affect the frequency with which markets clear?
- Does the presence of designated dealers result in more attractives prices posted in limit order books?

In limit order markets, traders can choose to place either a limit order or a market order. While the submission of a market order triggers an immediate execution at the limit prices posted in the book, a limit order faces a risk of non execution. Since market orders immediately generate a transaction while limit orders do not, the market clearing frequency is directly related to the order choice strategies of traders. Besides, to investigate how the presence of dealers affect prices, we need to characterize both quotes entered in the book by market makers and those posted by traders who choose to place a limit order. To answer the previous questions, we thus model a dynamic centralized limit order book with designated market makers. The model is constructed as an extension of Foucault (1999) in order to include designated market makers who are required to provide two-sided quotes in the book. Traders and market makers arrive sequentially to the marketplace. Traders can place either a market order or a limit order and, in the latter case, they can choose the order's limit price. Market makers are required to post both buy and sell limit orders in the book. We obtain a characterization of quotes and order placement strategies at equilibrium. Furthermore, comparison between a limit order market with market makers and a pure limit order market yields several testable implications.

First, the clearing frequency is higher in a limit order market with designated market makers. The presence of market makers entails a higher competition for the execution of limit orders. Indeed, a trader knows that if he opts for a limit order, his offer must be attractive enough so that a market maker cannot profitably improve upon his quote. Limit order trading thus turns

⁶The terms traders and investors are used interchangeably in this paper.

out to be less profitable in the presence of market makers. This entails that the submission of a market order is more often the optimal order choice strategy for traders. The model thus yields the empirical prediction that the regular presence of market makers should facilitate immediate transactions.

Since the added liquidity provided by market makers tends to deter traders from placing limit orders, the impact of the presence of market makers on price competition and the quotes posted in the order book is not straightforward. We show that the best quotes in the limit order book with designated market makers are either more or less attractive than in a pure order book depending on whether the best quote has been respectively placed by a trader or a market maker. Because traders face the competition of market makers for the execution of limit orders, they must place more attractive offers than in the absence of market makers. Furthermore, traders only submit limit orders when they know that market makers cannot improve upon their prices. Hence, a market maker's quote represents the best price on one side of the book when he is the only liquidity provider on that side. Since market makers do not have the same need to trade as final traders and are both potential sellers and buyers, their quotation strategy is less aggressive than that of traders in a pure limit order market.

In this context, a natural question to ask is whether the presence of market makers reduces transaction costs on an ex ante basis, that is before knowing whether the best quote has been set by a trader or a market maker. We find that expected best quotes in the order book can be more or less favorable in the presence of market makers depending on the asset volatility and the differences in valuation among traders. More precisely, other things equal, expected best prices in a hybrid limit order book become more attractive than in a pure limit order book when the asset volatility increases. The intuition is as follows. When the asset volatility increases, the losses due to the risk of being picked off are higher. Market makers thus shade more their offers. As a result, market order trading is more costly and the proportion of limit orders in the traders' order flow increases. The probability that the best quotes are set by traders (market makers) then increases (decreases). Since traders' limit prices improve upon the best quotes in a pure limit order book while the reverse is true for market makers' quotes, expected best prices in a limit order market with designated dealers become more attractive than in a pure order market.

This result yields empirical predictions relative to the impact of the presence of market

makers on expected prices in the book across assets and/or periods. Cross-sectional variations in the asset volatility or in the differences in valuation among traders should generate cross-sectional differences in the impact of the presence of designated dealers on expected prices. Since asset volatility decreases with equity capitalisation⁷, our results provide a rationale for the trading of mid caps being organized as limit order books with designated dealers. Another question of interest is related to the need of designated market makers on the large capitalisation stock markets. According to our results, the presence of market makers could well result in inferior prices if the asset volatility is small. Only their presence could have a positive impact on trading costs during periods of high volatility.

The impact of the presence of non monopolist dealers in limit order markets has recently become an empirical issue (Nimalendran and Petrella (2003), Mann, Venkataraman and Waisburd (2004), Anand, Tanggaard, Weaver (2005), Jain (2002)). Overall, these papers find that liquidity is enhanced following the entry of market makers in the limit order market^{8,9}. By contrast, we are not aware of theoretical models which investigate whether the presence of non monopolist market makers facilitates trading and reduces transaction costs in limit order markets. Theoretical models of limit order books either consider a pure limit order market (Glosten (1994), Biais, Martimort and Rochet (2000), Foucault (1999), Parlour (1998) etc) or model a hybrid specialist/limit order market which present the features of the NYSE (Glosten (1989), Seppi (1997), Parlour and Seppi (2003)). As argued above, the NYSE specialist is different from the

⁷See for instance Hasbrouck (1991).

⁸Nimalendran and Petrella (2003) find that different indicators of market quality are improved after market makers were introduced for thinly-traded stocks on the Italian Stock Exchange (ISE). Mann, Venkataraman and Waisburd (2004) focus on the impact of the non monopolist specialist for stocks traded in twice-daily call auctions on the Paris Bourse. Anand, Tanggaard, Weaver (2005) find that listed firms on the Stockholm Stock Exchange (SSE) benefit from increased liquidity and lower volatility following the entry of committed market makers. Note, however, that the SSE's trading system differ from the ISE and other exchanges which have adopted the hybrid limit order book analyzed in this paper since the SSE does not contract with market makers but allow firms to do so. By contrast, on Euronext, the ISE, the TSE or the LSE's SETSmm, market makers are directly accountable to the exchange. Some exchanges (Euronext or the ISE for instance) also allow listing firms to contract with market makers so that market making firms provide listed firms with additional services (financial analysis services, price support, etc) not required by the exchange. Jain (2002) finds that the presence of a designated market maker is associated with enhanced liquidity. However, this study does not distinguish between NYSE-based market structures and hybrid limit order markets with non monopolist market makers.

⁹A number of empirical papers analyze the behaviour of the NYSE hybrid specialist/limit order market (Kavajecz (1999), Chung, Van Ness and Van Ness (1999), Harris and Panchapagesan (2005) among others) or examine the value of the specialist in the options markets (Mayhew (2002), Anand and Weaver (2003), etc).

market makers we consider in this paper. Our analysis is motivated by features of Euronext, the LSE's SETSmm trading system for mid-caps or the Toronto Stock exchange, not by features of the NYSE. The NYSE specialist has an informational advantage over limit order traders in that he observes the realized size of arriving market orders before deciding whether and how to trade while limit order traders must post their offers before knowing the size of future market orders. By contrast, the market makers we consider have the same information than limit order traders regarding the future market order flow¹⁰ and must compete with other traders by posting offers in the order book. Thus none of NYSE-based models apply to the trading mechanism analyzed in this paper.

Besides, the models of Glosten (1989), Seppi (1997) and Parlour and Seppi (2003) and several models of pure limit order markets (Glosten (1994), Biais, Martimort and Rochet (2000), Viswanathan and Wang (2002)) are static. Thus they do not allow traders to optimally choose between a market order and a limit order and cannot investigate how the presence of dealers impacts the order choice strategies of traders and the clearing frequency. Dynamic models of limit order markets include Foucault (1999), Parlour (1998), Goettler, Parlour and Rajan (2003), Foucault, Kadan and Kandel (2004) and Rosu (2004). However, these papers consider pure limit order markets without designated market makers.

This paper proceeds as follows. The model is presented in section 2. In section 3, the optimal strategies of traders and market makers in a centralized limit order market are characterized. The benchmark case in which only traders participate to a pure limit order market is presented. Section 4 compares price formation and order placement strategies across a limit order market with market makers and a pure limit order market. Section 5 concludes. All proofs are collected in the appendix.

2 The Model

In this section, we present a dynamic model of a centralized limit order market in which two types of agents interact. Designated market makers provide continuous liquidity by posting buy and sell limit orders. Traders arrive sequentially and must decide to place either a limit or a

¹⁰For a more detailed comparison between the NYSE specialist and the Euronext market maker, see Mann, Venkataraman and Waisburd (2004). Charitou and Panayides (2004) offer a comprehensive descriptive comparison of different market structures which involve the presence of market makers.

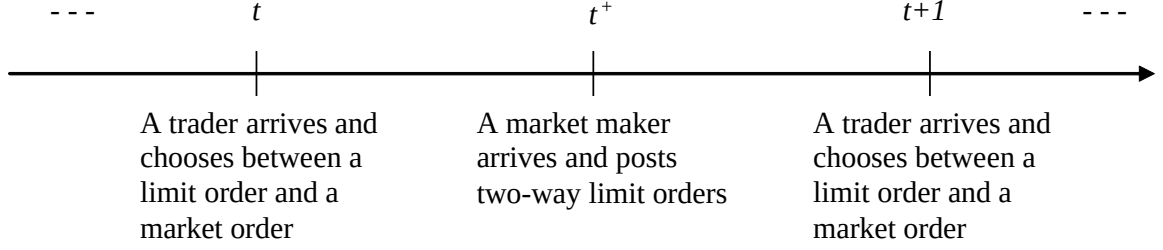


Figure 1: A sequence of events

market order.

The process of the asset value

We consider the market for a single risky asset whose expected value at time t is denoted by v_t . This value follows a random walk :

$$v_{t+1} = v_t + \varepsilon_{t+1}$$

where $\varepsilon = \pm\sigma$ with equal probabilities. The innovations reflect the fact that the asset value changes over time due to the arrival of new public information.

The model has an infinite horizon. Hence, we focus on stationary equilibria such that the agents' equilibrium strategies do not depend on the time at which they are decided. Consistent with this restriction, all exogenous parameters are assumed to be stationary.

Trading mechanism and timing

The trading process is sequential. At each time, a trader arrives in the market. He can be either a buyer or a seller and chooses between a market and a limit order. Between two traders' arrivals, a market maker posts a buy and a sell limit orders. Following Foucault (1999), it is assumed that all orders are for one unit of the asset and that limit orders expire after one period. Without designated market makers, these assumptions entail that the limit order book is either empty or full. Differently, the continuous presence of dealers ensures that the limit order book is never empty so that a trader always has the ability to submit a market order. The sequence of traders' and market makers' arrivals is depicted in Fig. ??.

At time t , the strategy of the trader consists of choosing between a market and a limit

order and, if he opts for a limit order, of determining his limit price. A market order executes immediately at the best offer in the book. In the presence of designated dealers, a market order submitted at time t can be matched with a limit order submitted either by a trader at time $(t - 1)$ or by a market maker at time $(t - 1)^+$. By contrast, a limit order faces a risk of non execution. Under our assumptions, a limit order submitted at time t is either executed at time $(t + 1)$ or expires. Hence, the probability of execution depends on the order placement strategies of the trader who arrives at $(t + 1)$ and of the market maker who arrives at time t^+ . The limit orders posted by the market maker at time t^+ are also held during one period. Since no trader arrives between times $(t + 1)$ and $(t + 2)$, the market maker's orders are either executed at time $(t + 1)$ or expire at time $(t + 1)^+$. In this latter case, the two-way quotes entered by the market maker at time t^+ are replaced by other market maker's offers at time $(t + 1)^+$.

Then, when choosing his order placement strategy, the trader must take into account not only the state of the book at the time he arrives on the market but also the way his order will affect the strategies of the market maker and the trader arriving next. In the presence of designated market makers, a trader who opts for a limit order will not be executed if the market maker who follows posts more aggressive offers in the book. How such competition for liquidity provision affects the order placement strategies and the best prices posted in the limit order book is the focus of this paper.

The trader and the market maker who arrive respectively at times t and t^+ are assumed to have the same information regarding the asset value. They both expect the asset value to be equal to v_t . Notice also that the market maker at time t^+ has the same information as the trader at time t over future order flow. For this reason, the market maker is not an information "monopolist" as characterized in Glosten (1989). Differently, at time t^+ the market maker observes the order placement strategy of the trader who arrived at time t .

Agents' private valuations

Agents differ by their reservation prices. For a buyer (seller), the reservation price is the highest (smallest) price the agent is willing to pay (receive) for the asset. At time τ , the reservation price of an agent who arrives at time t or t^+ is $v_\tau + r_t$, where r_t is the agent's private valuation for the asset.

Traders. A trader can be either a buyer or a seller with equal probabilities. A buyer attaches a higher private valuation to the asset than a seller. Hence, r_t can take two values with equal

probabilities $r_b = +L > 0$ for a buyer and $r_s = -L$ for a seller. They are several explanations for the dispersion of private valuations across traders. For instance, differences in valuation can result from heterogeneity across liquidity shocks faced by traders. As in most microstructure models, such dispersion is necessary to generate transactions.

Market makers. Market makers differ from traders in that they do not need to trade *per se* ; their main reason for posting limit orders is to meet their contractual obligations. Moreover, they must post two-way quotes in the book ; then, they are both potential sellers and buyers. Hence, it is natural to assume that market makers' private valuation r_m is both lower than buyers' valuation r_b and higher than sellers' one r_s . For simplicity, we assume that $r_m = 0$.

Preferences

Let $b_t = (\underline{A}_t, \overline{B}_t)$ denote the best quotes in the book at time t . When arriving to the market, agents observe their private valuation, the current asset value and the best quotes in the book. Since at time t^+ the market maker observes the order placement strategy of the trader who arrived at time t , the state of the book at time t , b_t , is different from the state b_{t+} at time t^+ . We denote by $s_t = (b_t, v_t)$ the public information on the book and the asset value available to a trader who chooses his strategy at time t .

All agents are assumed to maximize their expected utility and to be risk neutral. In the absence of a trade, the reservation utility of all the agents if they do not trade is normalized to zero.

Traders. If the trader who arrives at time t chooses to place a limit order at a price P , his expected utility is

$$U_l(s_t, r_t, P) = \phi \pi(P \mid s_t) [r_t + E(v_{t+1} \mid s_t, I(P)) - P]$$

where $\pi(P \mid s_t)$ is the probability of execution of the order, $I(P)$ is an indicator function which takes the value +1 in case of execution of the order and 0 otherwise and $\phi = +1$ ($\phi = -1$) if the trader submits a buy (sell) limit order. If the trader opts for a market order, then his expected utility is

$$U_m(s_t, r_t) = \phi [r_t + v_t - \underline{A}_t] + (1 - \phi) [\overline{B}_t - (r_t + v_t)]$$

When deciding whether to place a limit order or a market order, a trader who arrives at time t

must first determine his optimal limit price $P(v_t, r_t)$ if he opts for a limit order :

$$P(v_t, r_t) \in \arg \max_P U_l(s_t, r_t, P)$$

Then, the trader optimally places a limit order if $U_l(s_t, r_t, P(v_t, r_t))$ is superior to $U_m(s_t, r_t)$ and submits a market order otherwise. Moreover, it is assumed that a trader who is indifferent between a limit order and a market order chooses to trade immediately via a market order. Hereafter, the optimal ask and bid prices posted by respectively a seller and a buyer if they choose to place a limit order are denoted by $A(v_t, -L)$ and $B(v_t, +L)$.

Market makers. A market maker who arrives at time t^+ must submit both a buy and a sell limit orders. Let A^M and B^M denote the ask and bid prices he posts on the book. For tractability, we assume that each market maker arrives only once to the market. Thus, there is continuous liquidity provision by market makers in the order book but each market maker supplies liquidity only once. While this may not be the case in practice, notice that under our assumptions we focus on the best quotes provided in the book. Hence, as soon as they are several market makers¹¹, this assumption implies that the market maker who provides the best quotes among all market makers is different at each time. Notwithstanding this assumption, the model still captures the main feature of limit order books with non monopolist dealers, which is the regular liquidity provision by designated intermediaries¹².

Proceeding as for traders, the market maker who arrives at time t^+ has an expected utility defined by

$$\begin{aligned} U^M(s_{t+}, A^M, B^M) = & \pi(A^M | s_{t+}) [A^M - E(v_{t+1} | s_{t+}, I(A^M))] \\ & + \pi(B^M | s_{t+}) [E(v_{t+1} | s_{t+}, I(B^M)) - B^M] \end{aligned}$$

where $\pi(B^M | s_{t+})$ ($\pi(A^M | s_{t+})$) is the probability of execution of the buy (sell) limit order and $I(B^M)$ ($I(A^M)$) is an indicator function which takes the value +1 in case of execution of the

¹¹The Toronto Stock Exchange, The London Stock Exchange's SETSmm trading system and Euronext for many stocks allow several market makers to post orders in the limit order book.

¹²In practice, market makers are required to quote two-way prices in the order book, with a minimum volume size and within a maximum allowed bid-ask spread. However, the spread constraint is likely to be not binding in practice. For instance, the current maximum spread requirement for market makers who provide liquidity on the Euronext limit order book is 5%, unless for a stock included in the Next 150 index where the maximum spread is 4%. However, Euronext reports that 90% of equities have a spread strictly lower than 5% and 83.5% of equities have a spread strictly lower than 4%, the average spread for equities continuously traded being 2.11% (Source : Euronext zone equities 2004). Hence, we do not model in this paper the market makers' maximum spread requirements.

order and 0 otherwise. The first (second) term of the right-hand side of this expression is the market maker's expected utility relative to the ask (bid) side.

Buy and sell limit orders placed the market maker at time t^+ are either hit by the trader who arrives at time $(t + 1)$ or expire. Since this trader submits either a buy or a sell order, the ask and bid prices A^M and B^M posted by the market maker are independant. Hence, at time t^+ , the market maker determines his optimal prices $A^M(v_t)$ and $B^M(v_t)$ such that

$$\begin{aligned} A^M(v_t) &\in \arg \max_{A^M} U^M(s_{t+}, A^M) = \pi(A^M | s_{t+}) [A^M - E(v_{t+1} | s_{t+}, I(A^M))] \\ B^M(v_t) &\in \arg \max_{B^M} U^M(s_{t+}, B^M) = \pi(B^M | s_{t+}) [E(v_{t+1} | s_{t+}, I(B^M)) - B^M] \end{aligned}$$

Equilibrium definition

A trader's strategy involves the choice of (i) the type of order (market order versus limit order), and (ii) if he chooses a limit order, the limit price at which the order is placed. A market maker's strategy consists of the optimal ask and bid prices at which his limit orders are placed.

Notice that the expected utility of a buyer who submits a market order $U_m(s_t, +L)$ decreases with the best ask price offered in the book $\underline{A}_t$. Intuitively, there is a cutoff ask price such that, as soon as the best offer is higher than this cutoff value, the buyer prefers to submit a limit order rather than a market order. Let $C(v_t, +L)$ denote the highest ask price at which a buyer is willing to submit a market order. This cutoff value is such that

$$v_t + L - C(v_t, +L) = U_l(s_t, +L, B(v_t, +L)) \quad (1)$$

Symmetrically, the lowest bid price at which a seller is willing to submit a market order $C(v_t, -L)$ is given by

$$C(v_t, -L) - (v_t - L) = U_l(s_t, -L, A(v_t, -L)) \quad (2)$$

Definition 1 *An equilibrium is characterized by traders' order choice strategies $C(., +L)$ and $C(., -L)$, traders' optimal prices $B(., +L)$ and $A(., -L)$ if they opt for a limit order and market makers' optimal bid and offer prices $B^M(.)$ and $A^M(.)$ such that each agent's strategies maximize his expected utility given the strategies of other agents.*

3 Equilibrium in a Centralized Limit Order Market with Designated Market Makers

In this section, we characterize the equilibrium strategies of traders and market makers in a centralized limit order market. As a benchmark, we then describe the traders' optimal strategies that would prevail in a limit order market without market makers.

3.1 Price Formation and Order Placement Strategies in a Limit Order Market with Market Makers

Consider the problem faced by a market maker at time t^+ . He observes the order choice strategy of the last trader and, if the latter opts for a limit order, his quotation strategy. Let's suppose that this trader posts a buy limit order. Then, the market maker's bid limit price must be at least as attractive as the last trader's one if he wants his buy order to get a chance to be executed. Under our assumptions, any limit price posted beyond the best prices in the book have indeed a zero execution probability. In turn, at time t , the trader needs to conjecture how his decisions will affect the next market maker's quotation strategy. The choice of a limit order by the trader will yield him a strictly positive profit only if the next market maker cannot profitably improve upon his quote. Besides, both the trader's and market maker's choices depend on how their strategies will affect the order placement strategies of the trader who arrives at time $(t+1)$. The following proposition gives the optimal quotation strategies of agents when these determine the best prices in the book.

Proposition 1 *For all values of the parameters (L, σ) , there is a unique equilibrium of the limit order market with designated market makers.*

(i) *If $0 \leq \sigma < \frac{L}{4}$, traders only place market orders. The best prices are set by market makers and given by $A^M(v_t) = v_t - \sigma + \frac{L}{2}$ and $B^M(v_t) = v_t + \sigma - \frac{L}{2}$. The execution probability of these offers is equal to $1/2$.*

(ii) *If $\frac{L}{4} \leq \sigma < \frac{5L}{4}$, traders may submit limit or market orders. When the best prices are posted by traders, they are given by $A(v_t, -L) = v_t - \max\left\{0, \frac{(2\sigma-L)}{3}\right\}$ and $B(v_t, +L) = v_t + \max\left\{0, \frac{(2\sigma-L)}{3}\right\}$. The execution probability of these offers is equal to $1/2$. When the best*

prices are posted by market makers, they are given by $A^M(v_t) = v_t + \sigma + \frac{L}{2} + \max\left\{0, \frac{(2\sigma-L)}{6}\right\}$ and $B^M(v_t) = v_t - \sigma - \frac{L}{2} - \max\left\{0, \frac{(2\sigma-L)}{6}\right\}$. The execution probability of these offers is equal to $1/4$.

(iii) If $\frac{5L}{4} \leq \sigma$, traders may submit limit or market orders. When the best prices are posted by traders, they are given by $A(v_t, -L) = v_t + \sigma$ and $B(v_t, +L) = v_t - \sigma$. When the best prices are posted by market makers, they are given by $A^M(v_t) = v_t + \sigma + \frac{3L}{4}$ and $B^M(v_t) = v_t - \sigma - \frac{3L}{4}$. The execution probabilities of traders' and market makers' offers is equal to $1/4$.

This proposition identifies three regimes in terms of the market parameters (L, σ) . As in Foucault (1999), when σ increases and crosses a threshold ($\frac{L}{4}$ or $\frac{5L}{4}$), both traders and market makers shade their offers more : ask prices increase and bid prices decrease. This reflects the fact that when setting their limit prices, agents account for the risk of being picked off : limit orders are more likely to be executed when new information arrives and makes these orders become mispriced. This winner's curse problem induces agents to post less favorable quotes in the book when the asset volatility increases since the potential losses due to mispricing are higher in this case.

A trader always submits a market order when the asset volatility is small ($\sigma < \frac{L}{4}$). The intuition is as follows. When the asset volatility is low, the losses due to the risk of being picked off are small and market makers post attractive offers in the book. This results in market orders being more attractive than limit orders for traders. First, because offers in the book are favorable, market orders incur low transaction costs. Second, if a trader opts for a limit order, his offer must be such that the next market maker cannot profitably improve upon his quote. The trader's quote must then be so aggressive that the expected profit associated with a limit order is lower than with a market order. When the asset volatility increases, this result is no longer true. In this case, the winner's curse problem leads market makers to widen their spreads. Hence, the trading costs associated to the submission of market orders are higher and it can be more profitable for a trader to place a limit order than a market order.

The next two tables describe the optimal order choice strategy of a buyer (the case of a seller is symmetric) conditional on the strategy of the last trader for different values of the parameters¹³. When a trader opts for a buy (sell) limit order, the latter must be such that a

¹³The last step of the proof of Proposition 1 yields the results which appear in these tables.

market maker cannot profitably improve upon his quote. Hence, it represents the best quote on the buy (sell) side on the book. Differently, a marker maker' bid (ask) price is the best quote only if he is the only liquidity provider on the buy (sell) side of the order book. Otherwise, his price lies beyond the best quote. Any such price has a zero execution probability and yields a zero expected profit to the market maker. Hence, when the market maker cannot profitably improve upon the quote posted by the last trader, any market maker's price beyond the best quote is an equilibrium price. In the following table, $A_{ne}^M(.)$ denotes a possible equilibrium market maker's ask price when it is not the best ask price in the book.

$\frac{L}{4} \leq \sigma < \frac{5L}{4}$		Market maker's strategy at time $(t-1)^+$	Buyer's strategy at time t	
			$v_t = v_{t-1} + \sigma$	$v_t = v_{t-1} - \sigma$
Strategy of the trader at time $(t-1)$	Sell Limit order $A(v_{t-1}, -L)$	$A_{ne}^M(v_{t-1}) \geq A(v_{t-1}, -L)$ $B^M(v_{t-1})$	Market order	Market order
	Sell market order or buy order	$A^M(v_{t-1})$ $B^M(v_{t-1})$	Market order	Limit order $B(v_t, +L)$

Figure 2: Optimal order choice decisions of a buyer who arrives at time t and optimal market maker's limit prices placed at time $(t-1)^+$ conditional on the strategy of the last trader in a limit order market with designated market makers when $\frac{L}{4} \leq \sigma < \frac{5L}{4}$.

$\frac{5L}{4} \leq \sigma$		Market maker's strategy at time $(t-1)^+$	Buyer's strategy at time t	
			$v_t = v_{t-1} + \sigma$	$v_t = v_{t-1} - \sigma$
Strategy of the trader at time $(t-1)$	Sell Limit order $A(v_{t-1}, -L)$	$A_{ne}^M(v_{t-1}) \geq A(v_{t-1}, -L)$ $B^M(v_{t-1})$	Market order	Limit order $B(v_t, +L)$
	Sell market order or buy order	$A^M(v_{t-1})$ $B^M(v_{t-1})$		

Figure 3: Optimal order choice decisions of a buyer who arrives at time t and optimal market maker's limit prices placed at time $(t-1)^+$ conditional on the strategy of the last trader in a limit order market with designated market makers when $\frac{5L}{4} \leq \sigma$.

3.2 A Benchmark : a Limit Order Market without Designated Market Makers - Foucault (1999)

In order to determine how the presence of designated market makers affects price formation and order placement decisions in a limit order market, it is useful to consider a limit order market in which there is no market maker. Let $A^p(v_t, -L)$ and $B^p(v_t, +L)$ denote the optimal ask and bid prices posted by respectively a seller and a buyer in a pure limit order market. The following proposition follows directly from Proposition 3 in Foucault (99).

Proposition 2 *For all values of the parameters (L, σ) , there is a unique equilibrium of the limit order market without designated market makers.*

(i) *If $0 \leq \sigma < \frac{4L}{5}$, traders' optimal quotation strategies are given by $A^p(v_t, -L) = v_t - \frac{(2\sigma - L)}{3}$ and $B^p(v_t, +L) = v_t + \frac{(2\sigma - L)}{3}$. The execution probability of these offers is equal to $1/2$.*

(ii) *If $\frac{4L}{5} \leq \sigma$, traders' optimal quotation strategies are given by $A^p(v_t, -L) = v_t + \sigma + \frac{3}{5}L$ and $B^p(v_t, +L) = v_t - \sigma - \frac{3}{5}L$. The execution probability of these offers is equal to $1/4$.*

When the asset volatility crosses the threshold $\frac{4L}{5}$, the offers in the book become less attractive. It follows that market order trading is more costly. Thus more traders prefer to place a limit order rather than a market order and the execution probability of limit orders decrease. As a way to facilitate the comparison between the two market designs under analysis in this paper, the following tables describe the optimal order choice strategy of a buyer conditional on the strategy of the last trader for different values of the parameters in a pure limit order market.

4 Do Market Makers Improve Liquidity in Limit Order Markets?

Using the results in Propositions 1 and 2, this section investigates how the presence of market makers in a limit order market impacts the mix between market orders and limit orders in the order flow and the trading costs incurred by market order submitters. Several empirical hypothesis for the value of market makers are derived from the comparison of a limit order market with and without designated market makers.

$0 \leq \sigma < \frac{4L}{5}$		Buyer's strategy at time t	
		$v_t = v_{t-1} + \sigma$	$v_t = v_{t-1} - \sigma$
Strategy of the trader at time ($t-1$)	Sell Limit Order $A^p(v_{t-1}, -L)$	Market order	Market order
	Sell market order or buy order	Limit order $B^p(v_t, +L)$	Limit order $B^p(v_t, +L)$

Figure 4: Optimal order choice decisions of a buyer who arrives at time t conditional on the strategy of the last trader in a limit order market without designated market makers when $0 \leq \sigma < \frac{4L}{5}$.

$\frac{4L}{5} \leq \sigma$		Buyer's strategy at time t	
		$v_t = v_{t-1} + \sigma$	$v_t = v_{t-1} - \sigma$
Strategy of the trader at time ($t-1$)	Sell Limit Order $A^p(v_{t-1}, -L)$	Market order	Limit order $B^p(v_t, +L)$
	Sell market order or buy order	Limit order $B^p(v_t, +L)$	Limit order $B^p(v_t, +L)$

Figure 5: Optimal order choice decisions of a buyer who arrives at time t conditional on the strategy of the last trader in a limit order market without designated market makers when $\frac{4L}{5} \leq \sigma$.

4.1 Order Flow Composition

Does the presence of market makers affect the type of orders chosen by investors? Does the added liquidity provided by market makers enhance the competition for liquidity supply and foster the submission of limit orders? Or does it facilitate immediate transactions, thereby deterring traders from submitting limit orders? These questions have both a theoretical and an empirical interests for researchers, firms and market designers.

Let $I_i(MO)$ ($I_i(LO)$) be an indicator function which takes the value +1 if the trader who arrives at time i submits a market order (limit order) and 0 otherwise. The total number of

traders' orders (limit and market orders) in the interval of time $[0, (t - 2)]$ is $(t - 1)$. Hence, the expected proportion of market orders in the order flow in this interval of time is

$$m_t = \frac{1}{(t - 1)} \sum_{i=0}^{(t-2)} E(I_i(MO))$$

Since we focus in this paper on stationary equilibria, the proportion of market orders in the order flow at equilibrium is defined by $m = \lim_{t \rightarrow +\infty} m_t$. The proportion of limit orders at equilibrium is then $(1 - m)$. The following propositions give the equilibrium proportion of market orders in the traders' order flow in both a limit order market with designated market makers and without them. Proposition 4 follows directly from Proposition 4 in Foucault (99).

Proposition 3 *At equilibrium in a limit order market with designated market makers, the proportion of market orders in the traders' order flow is (i) 100% if $0 \leq \sigma < \frac{L}{4}$, (ii) 3/5 if $\frac{L}{4} \leq \sigma < \frac{5L}{4}$ and (iii) 50% if $\sigma \geq \frac{5L}{4}$.*

In both market designs, the proportion of market orders is negatively related to asset volatility. This result follows directly from the winner's curse problem faced by limit order submitters. As the asset becomes more volatile, the losses associated to the risk for limit order submitters of being executed in case of an unfavorable evolution of the asset are higher. Hence, limit order traders charge higher ask prices and lower bid prices to market order submitters. Market order trading results less attractive and traders' optimal order choice strategy consists more frequently in submitting limit orders.

Proposition 4 *At equilibrium in a limit order market without designated market makers, the proportion of market orders in the order flow is (i) 1/3 if $0 \leq \sigma < \frac{4L}{5}$ and (ii) 1/5 if $\sigma \geq \frac{4L}{5}$.*

Besides, the comparison of both market designs shows that the equilibrium proportion of market orders in the order flow is higher when designated market makers are present in the limit order market than when they are not. The intuition is as follows. The presence of market makers entails a higher competition for the execution of limit orders. A trader knows that if he opts for a limit order, his offer must be attractive enough so that the next market maker cannot profitably improve upon his quote. Limit order trading thus turns out to be less profitable in the presence of market makers. This entails that the submission of a market order is more often the

optimal order choice strategy for traders. Moreover, the continuous presence of market makers implies that the submission of a market order is always possible, while this is not the case in a pure limit order market.

Corollary 1 *The proportion of market orders in the order flow is higher in a limit order market with designated market makers than in a pure limit order market.*

Since market orders immediately generate a transaction and limit orders do not, this result yields the empirical prediction that the presence of designated market makers in a centralized limit order market should increase the frequency with which the market clears. This is consistent with the findings in Venkataraman, Waisburd, Mann (2004) who study the value of the non-monopolist specialist on the Paris Bourse. However, their focus is on stocks that trade in twice-daily call auctions. It still remains an open question whether the presence of designated dealers leads to a higher clearing frequency in continuous limit order markets.

The direct corollary of this result is that the presence of designated market makers does not stimulate the provision of liquidity by final traders. The added liquidity provided by market makers tends to deter traders from placing limit orders. In a more general framework, the overall effect of the presence of market makers on market depth on each side of the book should be ambiguous.

- *Hypothesis 1 : The clearing frequency is higher in a centralised limit order market with designated market makers than in a pure limit order market.*

4.2 Best Offers in Limit Order Markets

The last corollary yields that the presence of market makers induces traders to opt less frequently for limit orders. Thus, a natural question to ask is whether the price competition and the resulting best offers in the book are enhanced in the presence of market makers or not. The following proposition compares the best prices posted in a limit order market either with designated market makers or without.

Proposition 5 *Comparison of best prices in a limit order market with designated market makers and without them yields the following results.*

(i) *When the best prices are posted by traders in the limit order book with market makers, the ask and bid prices are respectively lower and higher than in a limit order market without market makers : $A(v_t, -L) \leq A^p(v_t, -L)$ and $B(v_t, +L) \geq B^p(v_t, +L)$.*

(ii) *When the best prices are posted by market makers, the ask and bid prices are respectively higher and lower than in a limit order market without market makers : $A^M(v_t) \geq A^p(v_t, -L)$ and $B^M(v_t) \leq B^p(v_t, +L)$.*

In the limit order market with designated market makers, traders only submit limit orders when they know that market makers cannot improve upon their prices. Hence, a market maker's quote represents the best price on one side of the book if he is the only liquidity provider on that side. Since market makers do not have the same need to trade as final traders and are both potential sellers and buyers, their private valuation r_m is both lower than buyers' value r_b and higher than sellers' ones r_s . This entails that the quotation strategy of market makers is less aggressive than that of traders in a pure limit order market. By contrast, traders who face the competition of market makers for the execution of limit orders must place more attractive offers than in the absence of market makers. This results from the market makers' ability to undercut or outbid the quotes previously placed in the book by traders. This proposition yields the empirical prediction that trading costs incurred by market order submitters in a limit order market with market makers may actually be higher or lower than in a pure limit order market depending on whether the best quote has respectively been placed by a market maker or a limit order trader.

- *Hypothesis 2 : The best quotes in a limit order market with designated market makers are less attractive than in a pure limit order market when the best quotes are placed by market makers.*
- *Hypothesis 3 : The best quotes in a limit order market with designated market makers are more attractive than in a pure limit order market when the best quotes are placed by investors.*

Proposition 5 does not yield unambiguous results relative to how the presence of designated dealers impacts the best prices in the order book. Then, we may wonder whether the presence of market makers reduces transaction costs on an ex ante basis, that is before knowing whether

the best quote has been set by a trader or a market maker. More precisely, consider a seller who arrives to the marketplace at time t . Before observing the book, the best expected bid price at time t is defined by

$$B(v_{t-1}) = p_{t-1}(BuyLO) B(v_{t-1}, +L) + [1 - p_{t-1}(BuyLO)] B^M(v_{t-1}) \quad (3)$$

where $p_{t-1}(BuyLO)$ (resp. $(1 - p_{t-1}(BuyLO))$) is the probability that the best bid price in the book just before time t has been set by the trader (resp. market maker) who arrived at time $(t-1)$ ($(t-1)^+$). Since a trader only submits a limit order when it is not profitable for the next market maker to improve upon his price, $p_{t-1}(BuyLO)$ is thus equal to the probability that the trader at time $(t-1)$ submits a buy limit order. Moreover, at the stationary equilibrium, $p_{t-1}(BuyLO)$ does not depend on t and is equal to the proportion of buy limit orders in the order flow. Using Propositions 1 and 3, it is then possible to compute the expected best bid and ask prices in the book. These prices offer a way to compare expected transaction costs associated with the submission of a market order across a limit order book with designated dealers and a pure limit order book. In a limit order book without designated market makers, the expected best price is simply the same as the best price given by either $A^p(v_t, -L)$ or $B^p(v_t, +L)$.

Proposition 6 *Comparison of expected best prices in a limit order market with designated market makers and a pure limit order market yields the following results.*

(i) *If $0 \leq \sigma < \frac{4L}{5}$, expected best ask (bid) prices are higher (lower) in a limit order market with designated market makers than in a pure limit order market.*

(ii) *If $\frac{4L}{5} \leq \sigma$, expected best ask (bid) prices are lower (higher) in a limit order market with designated market makers than in a pure limit order market.*

Given a level L of the dispersion of traders' private valuations, expected best quotes in the order book can be more or less favorable in the presence of market makers depending on the asset volatility. The reason for this result is the following. When the asset volatility increases, the proportion of market orders in the order flow decreases. In other words, limit order trading turns out to be the traders' optimal order choice strategy more frequently. As a result, the probability that the best quotes are set by traders (market makers) increases (decreases). Since traders' limit prices improve upon the best quotes in a pure limit order book while the reverse is true for market makers' quotes, the expected best prices posted in a hybrid limit order book become more attractive than in a pure limit order book.

Interestingly, the presence of market makers can result in more attractive offers in the limit order book although the market makers' quotes are always inferior to those posted in a pure limit order market. At the root of the expected price improvement is the competitive pressure exerted by market makers on limit order submitters. The latter must account for the fact that their limit orders will not execute as soon as market makers improve upon their quotes. This induces them to adopt more aggressive pricing strategies.

This result yields empirical predictions relative to the impact of the presence of market makers on expected prices in the book across assets and/or periods. Cross-sectional variations in the asset volatility or in the differences in valuation among traders should generate cross-sectional differences in the impact of the presence of designated dealers on expected prices. Most of limit order books in which the provision of liquidity is supported by two-way quotes by designated non monopolist market makers have been implemented for the trading of mid caps. Examples include the Euronext limit order book where liquidity provider agreements are only allowed on small and mid caps or the London Stock Exchange's SETSmm trading system for mid-caps. Since asset volatility decreases with equity capitalisation, our results provide a rationale for the trading of mid caps being organized as limit order books with designated dealers. Another question of interest is related to the need of designated market makers on the large capitalisation stock markets. According to our results, the presence of market makers could well result in inferior prices if the asset volatility is small. Only their presence could have a positive impact on trading costs during periods of high volatility. This result is consistent with the findings in Nimalendran and Petrella (2003). The authors compare a pure limit order market with a hybrid limit order market with market makers for thinly traded stocks. They find that very thinly traded stocks benefit more from the adoption of a hybrid limit order market than more actively traded stocks in terms of liquidity.

- *Hypothesis 4 : The presence of designated market makers in limit order markets can result in more or less favorable expected best quotes depending on the asset volatility and the differences in valuation among traders.*

Notice that we have not considered in this paper the maximum spread requirements which market makers face since these constraints are likely to be not binding in practice (see footnote n° 12 p.9). Independant on this fact, the last result shows that while such requirements need not be necessary for some parameters' values ($\frac{4L}{5} \leq \sigma$), they become important when the presence

of market makers can result in inferior prices. In this latter case, only an appropriate maximum spread requirement could mitigate the negative impact of the presence of market makers on expected prices. More generally, an optimal maximum spread constraint needs not be constant across assets and/or over time.

5 Conclusion

In this paper, we present a dynamic model of a centralized limit order market with designated non monopolist market makers. Traders and market makers arrive sequentially to the marketplace. Traders can place either a market order or a limit order and, in the latter case, they can choose the order's limit price. Market makers are required to provide two-sided quotes in the book. We obtain a characterization of quotes and order placement strategies at equilibrium. Furthermore, comparison between a limit order market with market makers and a pure limit order market yields the following testable implications.

H1 : The clearing frequency is higher in a limit order market with designated market makers.

H2 : The best quotes in a limit order market with designated market makers are less attractive than in a pure limit order market when the best quotes are placed by market makers.

H3 : The best quotes in a limit order market with designated market makers are more attractive than in a pure limit order market when the best quotes are placed by investors.

H4 : On an ex ante basis, the presence of market makers in limit order markets can result in more or less favorable expected best quotes depending on the asset volatility and the differences in valuation among traders.

The presence of market makers thus facilitates immediate transactions but has a more ambiguous effect on the attractiveness of quotes posted in the book. The competition between market makers and traders for the execution of limit orders is at the root of these results. In equilibrium, traders who choose to place a limit order adopt an aggressive quotation strategy so that market makers cannot profitably improve upon their quotes. It follows that limit order trading turns out to be less profitable and traders place market orders more frequently in the presence of designated dealers. Furthermore, as the asset volatility increases, the proportion of limit orders in the traders' order flow increases. Thus, the more favorable quotes posted by

limit order traders in the presence of market makers determine more frequently the best quotes in the book. For this reason, as soon as the asset volatility is high enough, expected prices are more attractive in a limit order market with designated dealers than in a pure limit order market. Since asset volatility decreases with equity capitalisation, these results offer a rationale for the decision taken by an increasing number of stock exchanges to adopt such hybrid limit order books for the trading of mid caps. Our analysis indeed predicts that small firms should benefit from the presence of non monopolist market makers while this may not be the case for big firms.

In order to obtain analytical results, we have assumed that orders are for one unit and expire after one period. Thus, one extension of this paper consists of relaxing these assumptions and studying how the presence of non monopolist market makers affects the market depth in a limit order book. Another interesting research avenue would be to investigate how the number of designated market makers impacts their overall effect on market liquidity. These exercises are left for future research.

6 Appendix

Proof of Proposition 1

The proof first characterizes a market maker's optimal quotation strategy, when the strategies of the last and next traders are taken as given. Second, the optimal limit price posted by a trader if he opts for a limit order is determined. Third, we characterize traders' optimal order choice strategies. In what follows, optimal strategies are derived under the conjecture that cutoff prices $C(v_t, +L)$ and $C(v_t, -L)$ are increasing in the asset value v_t . Then it is verified that this conjecture is indeed correct.

- *Step 1 : Characterization of market makers' optimal responses to traders' strategies*

Consider a market maker who arrives at time t^+ and let $C(., +L)$ and $C(., -L)$ be possible candidates for the traders' equilibrium order choice strategies. Before he chooses his optimal bid and ask prices, the market maker observes the limit order book. The latter can either be empty or contain one limit price entered by the trader who has arrived at time t . Consider the

decision to place his optimal bid price $B^M(v_t)$. Just before time t^+ , it is possible to distinguish between three possible states of the bid side of the order book.

Case 1 *there is a buy limit order at a price B such that $B \geq C(v_t + \sigma, -L) > C(v_t - \sigma, -L)$*

In this case, as soon as the market maker places his bid price below B , the execution probability of his buy order and the associated expected utility are equal to zero. The only way for the market maker to get executed is then to post a bid price $B^M(v_t)$ higher than B . Moreover every price higher than B has the same execution probability, which is the probability that a seller arrives at time $(t + 1)$. Indeed, as soon as a seller faces a bid price higher than $C(v_t + \sigma, -L)$, he submits a market order whether the asset value increases ($v_{t+1} = v_t + \sigma$) or decreases ($v_{t+1} = v_t - \sigma$). Hence, the market maker optimally posts a bid price slightly higher than B and gets the expected utility $U^M(\text{Case1}, B) = \frac{1}{2} [v_t - B]$ as soon as $B < v_t$. If $B \geq v_t$, then the market maker cannot profitably outbid the quote placed by the trader at time t and any bid price lower than B is an equilibrium quotation strategy which yields a zero expected utility to the market maker.

Case 2 *there is a buy limit order at a price B such that $C(v_t + \sigma, -L) > B \geq C(v_t - \sigma, -L)$*

In this case, the market maker must still outbid the trader's bid price B to get a chance to be executed. Besides, he can now place his quote $B^M(v_t)$ either in the interval $]B, C(v_t + \sigma, -L)[$ or $[C(v_t + \sigma, -L), +\infty[$. Any bid price in the interval $]B, C(v_t + \sigma, -L)[$ has the same execution probability. It is executed only if the asset value decreases ($v_{t+1} = v_t - \sigma$) and a seller arrives at time $(t + 1)$. Hence, if the market maker chooses a bid price in the interval $]B, C(v_t + \sigma, -L)[$, he optimally posts the smallest bid price in this interval, that is a bid price slightly higher than B . In this case, he gets an expected utility $U^M(\text{Case2}, B) = \frac{1}{4} [v_t - \sigma - B]$. Such a strategy is profitable iff $B < v_t - \sigma$.

Similarly, any bid price $B^M(v_t)$ higher than $C(v_t + \sigma, -L)$ has the same execution probability of $1/2$, the probability that a seller arrives. Hence, if the market maker chooses a bid price in the interval $[C(v_t + \sigma, -L), +\infty[$, he optimally posts a bid price equal to $C(v_t + \sigma, -L)$ and obtains the expected utility $U^M(\text{Case2}, C(v_t + \sigma, -L)) = \frac{1}{2} [v_t - C(v_t + \sigma, -L)]$. This strategy is profitable iff $C(v_t + \sigma, -L) < v_t$.

If both prices B and $C(v_t + \sigma, -L)$ yield a negative expected utility, then the market maker cannot profitably outbid the quote placed by the trader who arrived at time t and any bid price lower than B is an equilibrium quotation strategy which yields zero expected utility to the market maker. If one of these two quotation strategies is profitable and the other one is not, the market maker chooses to post the bid price which yields a positive expected utility. If both prices B and $C(v_t + \sigma, -L)$ yield a positive expected utility to the market maker, he chooses to place a bid price slightly higher than B iff $U^M(\text{Case2}, B) > U^M(\text{Case2}, C(v_t + \sigma, -L))$ and chooses $C(v_t + \sigma, -L)$ otherwise. After rewriting the last inequality, the market maker chooses a bid price slightly higher than B iff $C(v_t + \sigma, -L) > \frac{1}{2}[v_t + \sigma + B]$ and $C(v_t + \sigma, -L)$ otherwise.

Case 3 *the buy side of the limit order book is empty*

In this case, the market maker is the only liquidity provider on the buy side. Proceeding as in Case 2, any bid price in the interval $[C(v_t - \sigma, -L), C(v_t + \sigma, -L)[$ has the same execution probability of $1/4$. If the market maker posts a price in this interval, he then optimally chooses a bid price equal to $C(v_t - \sigma, -L)$ and obtain an expected utility equal to $U^M(\text{Case3}, C(v_t - \sigma, -L)) = \frac{1}{4}[v_t - \sigma - C(v_t - \sigma, -L)]$. This strategy is profitable iff $C(v_t - \sigma, -L)$ is strictly inferior to $(v_t - \sigma)$.

As in Case 2, the market maker may also place a buy order at a price equal to $C(v_t + \sigma, -L)$ and obtain an expected utility of $U^M(\text{Case3}, C(v_t + \sigma, -L)) = \frac{1}{2}[v_t - C(v_t + \sigma, -L)]$.

If both pricing strategies yield a positive expected utility, he chooses a bid price equal to $C(v_t - \sigma, -L)$ iff $U^M(\text{Case3}, C(v_t - \sigma, -L)) > U^M(\text{Case3}, C(v_t + \sigma, -L))$ and chooses $C(v_t + \sigma, -L)$ otherwise. After rewriting the last inequality, the market maker chooses a bid price equal to $C(v_t - \sigma, -L)$ iff $C(v_t + \sigma, -L) > \frac{1}{2}[v_t + \sigma + C(v_t - \sigma, -L)]$ and $C(v_t + \sigma, -L)$ otherwise.

Notice that it is never optimal for the trader at time t as for the market maker at time t^+ to post a bid price strictly lower than $C(v_t - \sigma, -L)$. Indeed, at time $(t+1)$, a seller submits a sell market order iff the best bid price in the book is higher than or equal to $C(v_{t+1}, -L)$. Since $v_{t+1} = v_t \pm \sigma$, the execution probability of a buy limit order at a price $P < C(v_t - \sigma, -L) < C(v_t + \sigma, -L)$ is equal to zero.

Since the buy and sell sides are symmetric in this model, a symmetrical reasoning can be made to determine the market maker optimal ask price $A^M(v_t)$ at time t^+ . For brevity, we only

report the possible optimal ask prices posted by the market maker. Just before time t^+ , there are three possible states of the ask side of the order book.

Case 4 *there is a sell limit order at a price A such that $A \leq C(v_t - \sigma, +L) < C(v_t + \sigma, +L)$*

In this case, the market maker must undercut the trader's price to get a chance to be executed. He optimally posts an ask price $A^M(v_t)$ slightly lower than A and gets the expected utility $U^M(\text{Case4}, A) = \frac{1}{2} [A - v_t]$ as soon as $A > v_t$. If $A \leq v_t$, the market maker cannot profitably undercut the quote placed by the trader who arrived at time t and any ask price higher than A is an equilibrium quotation strategy which yields a zero expected utility to the market maker.

Case 5 *there is a sell limit order at a price A such that $C(v_t - \sigma, +L) < A \leq C(v_t + \sigma, +L)$*

In this case, the market maker may post an ask price either slightly lower than A or equal to $C(v_t - \sigma, +L)$. In the former case, he gets an expected utility equal to $U^M(\text{Case5}, A) = \frac{1}{4} [A - v_t - \sigma]$ which is strictly positive iff $A > v_t + \sigma$. In the latter case, his expected utility writes $U^M(\text{Case5}, C(v_t - \sigma, +L)) = \frac{1}{2} [C(v_t - \sigma, +L) - v_t]$ and is profitable iff $C(v_t - \sigma, +L) > v_t$. If both pricing strategies yield a positive expected utility, the market maker chooses an ask price slightly lower than A iff $U^M(\text{Case5}, A) > U^M(\text{Case5}, C(v_t - \sigma, +L))$ and chooses $C(v_t - \sigma, +L)$ otherwise. After rewriting the last inequality, the market maker chooses an ask price slightly lower than A iff $C(v_t - \sigma, +L) < \frac{1}{2} [A + v_t - \sigma]$ and $C(v_t - \sigma, +L)$ otherwise. If both prices A and $C(v_t - \sigma, +L)$ yield a negative expected utility, then the market maker cannot profitably undercut the quote placed by the trader who arrived at time t and any ask price higher than A is an equilibrium quotation strategy which yields zero expected utility to the market maker.

Case 6 *the sell side of the limit order book is empty*

In this case, the market maker may post an ask price equal either to $C(v_t + \sigma, +L)$ or to $C(v_t - \sigma, +L)$. In the former case, he gets an expected utility equal to $U^M(\text{Case6}, C(v_t + \sigma, +L)) = \frac{1}{4} [C(v_t + \sigma, +L) - v_t - \sigma]$ which is strictly positive iff $C(v_t + \sigma, +L) > v_t + \sigma$. In the latter case, his expected utility is $U^M(\text{Case6}, C(v_t - \sigma, +L)) = \frac{1}{2} [C(v_t - \sigma, +L) - v_t]$. If both pricing strategies yield a positive expected utility, the market maker chooses an ask price equal to

$C(v_t + \sigma, +L)$ iff $U^M(\text{Case6}, C(v_t + \sigma, +L)) > U^M(\text{Case6}, C(v_t - \sigma, +L))$ and chooses $C(v_t - \sigma, +L)$ otherwise. After rewriting the last inequality, the market maker chooses an ask price equal to $C(v_t + \sigma, +L)$ iff $C(v_t - \sigma, +L) < \frac{1}{2} [C(v_t + \sigma, +L) + v_t - \sigma]$ and $C(v_t - \sigma, +L)$ otherwise.

- *Step 2 : Optimal bid and offer prices set by traders if they opt for a limit order*

Consider a buyer who arrives to the marketplace at time t . If he places a buy limit order, he must account for the best response of the market maker at time t^+ . Indeed, if the market maker outbids the trader's limit price, then the trader has a zero probability to get executed and his expected utility is zero. The trader then only places a limit order if it has a strictly positive execution probability, ie if the market maker cannot profitably outbid. The buyer's limit price at t also depends upon the sellers' order choice strategy $C(., -L)$ and then on the sellers' optimal quotation strategies. Indeed, for the buyer's order to be executed, the next trader must be a seller and he must choose a market order. Hence, we need to determine buyers' and sellers' strategies simultaneously.

If a buyer submits a limit order, his limit price B can either be such that $B \geq C(v_t + \sigma, -L) > C(v_t - \sigma, -L)$ (Case 1) or $C(v_t + \sigma, -L) > B \geq C(v_t - \sigma, -L)$ (Case 2). Symmetrically, if a seller submits a limit order, his limit price A can either be such that $A \leq C(v_t - \sigma, +L) < C(v_t + \sigma, +L)$ (Case 4) or $C(v_t - \sigma, +L) < A \leq C(v_t + \sigma, +L)$ (Case 5). These cases enable to identify two different regimes for the traders' quotation strategies : in Cases 1 and 4, the traders' quotation strategies are *aggressive* while they are *weakly aggressive* in Cases 2 and 5.

We first characterize buyers' and sellers' optimal quotations strategies by assuming that each category of traders adopts either an aggressive or a weakly aggressive quotation strategy. Then, we compare the expected utilities obtained by traders depending on the aggressiveness of their strategy and we determine their optimal quotation strategies.

1. Characterization of traders' optimal quotation strategies when all traders follow an aggressive strategy (Cases 1 and 4)

In these cases, when the asset value is v_t , the buyer's optimal price is the smallest bid price which verifies $B \geq C(v_t + \sigma, -L)$ such that the market maker cannot profitably outbid. Step 1 shows that this is the case iff $B \geq v_t$. Without loss of generality, we can write $B = v_t + u$. The

buyer's program then writes

$$u_{Cases1\&4} = \min \{u; u \geq 0, v_t + u \geq C_{Cases1\&4}(v_t + \sigma, -L), u \leq L\} \quad (4)$$

The last inequality $u \leq L$ reflects the fact that the optimal bid price must yield a positive expected profit to the trader (so the bid price must be lower than his reservation value $v_t + L$). In Case 1, the buyer's execution probability is the probability that a seller arrives at time $(t+1)$. Hence, his expected profit writes $U_l(Case1, +L, u) = \frac{1}{2} [L - u]$. Then, using eq. 1, the buy cutoff price $C(v_t, +L)$ is determined by

$$C_{Case1}(v_t, +L) = v_t + \frac{(L + u)}{2} \quad (5)$$

Proceeding symmetrically for a seller, in Case 4 when the asset value is v_t , the seller's optimal price is the highest ask price which verifies $A \leq C(v_t - \sigma, +L)$ such that the market maker cannot profitably undercut. Step 1 shows that this is the case iff $A \leq v_t$. If we write $A = v_t - t$, the seller's program is

$$t_{Cases1\&4} = \min \{t; t \geq 0, v_t - t \leq C_{Cases1\&4}(v_t - \sigma, +L), t \leq L\} \quad (6)$$

When he places a sell limit order at a price A , the seller's expected profit is given by $U_l(Case4, -L, t) = \frac{1}{2} [L - t]$ and his sell cutoff price $C(v_t, -L)$ defined in eq. 2 is

$$C_{Case4}(v_t, -L) = v_t - \frac{(L + t)}{2} \quad (7)$$

Using eq. 5 and 7, the conditions $v_t + u \geq C_{Cases1\&4}(v_t + \sigma, -L)$ and $v_t - t \leq C_{Cases1\&4}(v_t - \sigma, +L)$

respectively rewrite as

$$2u + t \geq 2\sigma - L$$

and

$$u + 2t \geq 2\sigma - L$$

The solutions $u_{Cases1\&4}$ and $t_{Cases1\&4}$ to programs 4 and 6 are then given by $u_{Cases1\&4} = \max \{0, \frac{1}{2} (2\sigma - L - t_{Cases1\&4})\}$ and $t_{Cases1\&4} = \max \{0, \frac{1}{2} (2\sigma - L - u_{Cases1\&4})\}$ if $u_{Cases1\&4} \leq L$ and $t_{Cases1\&4} \leq L$. These solutions rewrite as $u_{Cases1\&4} = t_{Cases1\&4} = \max \left\{0, \frac{(2\sigma - L)}{3}\right\}$. Hence, if traders place limit orders and follow an aggressive quotation strategy, their optimal bid and offer prices are given by

$$B_{Case1\&4}(v_t, +L) = v_t + \max \left\{0, \frac{(2\sigma - L)}{3}\right\} \quad (8)$$

and

$$A_{Case1\&4}(v_t, -L) = v_t - \max \left\{ 0, \frac{(2\sigma - L)}{3} \right\} \quad (9)$$

Traders' expected profit is the same for buyers and sellers and writes

$$U_l(Cases1\&4, +L) = U_l(Cases1\&4, -L) = \frac{1}{2} \left[L - \max \left\{ 0, \frac{(2\sigma - L)}{3} \right\} \right]$$

Such aggressive strategies are profitable iff $u_{Cases1\&4} \leq L$ and $t_{Cases1\&4} \leq L$, which also write

$$\sigma \leq 2L$$

Traders cutoff prices are given by

$$C_{Case1\&4}(v_t, +L) = v_t + \frac{L}{2} + \max \left\{ 0, \frac{(2\sigma - L)}{6} \right\} \quad (10)$$

$$C_{Case1\&4}(v_t, -L) = v_t - \frac{L}{2} - \max \left\{ 0, \frac{(2\sigma - L)}{6} \right\} \quad (11)$$

2. Characterization of traders' optimal quotation strategies when all traders follow a weakly aggressive strategy (Cases 2 and 5)

The proof proceeds in a similar way. In cases 2 and 5, when the asset value is v_t , the buyer's optimal price is the smallest bid price which verifies $C(v_t + \sigma, -L) > B \geq C(v_t - \sigma, -L)$ such that the market maker cannot profitably outbid. Step 1 shows that this is the case iff $B \geq v_t - \sigma$ and $C(v_t + \sigma, -L) \geq v_t$. Without loss of generality, we can write $B = v_t - \sigma + u$. The buyer's program then writes

$$u_{Cases2\&5} = \min u \quad \text{such that} \quad (12)$$

$$L \geq u \geq 0 \quad (13)$$

$$C_{Cases2\&5}(v_t + \sigma, -L) > v_t - \sigma + u \geq C_{Cases2\&5}(v_t - \sigma, -L) \quad (14)$$

$$C_{Cases2\&5}(v_t + \sigma, -L) \geq v_t \quad (15)$$

In Case 2, the buyer's limit order is executed only if the asset value decreases ($v_{t+1} = v_t - \sigma$) and if a seller arrives at time $(t+1)$. Hence, his expected profit writes $U_l(Case2, +L, u) = \frac{1}{4} [L - u]$. Then, using eq. 1, the buy cutoff price $C(v_t, +L)$ is determined by

$$C_{Case2}(v_t, +L) = v_t + \frac{(3L + u)}{4} \quad (16)$$

Proceeding symmetrically for a seller, we can write $A = v_t + \sigma - t$ and the seller's program is

$$t_{Cases2\&5} = \min t \quad \text{such that} \quad (17)$$

$$L \geq t \geq 0 \quad (18)$$

$$C_{Cases2\&5}(v_t + \sigma, +L) \geq v_t + \sigma - t > C_{Cases2\&5}(v_t - \sigma, +L) \quad (19)$$

$$C_{Cases2\&5}(v_t - \sigma, +L) \leq v_t \quad (20)$$

The seller's expected profit is given by $U_l(Case5, +L, t) = \frac{1}{4}[L - t]$ and his sell cutoff price $C(v_t, -L)$ defined in eq. 2 is

$$C_{Case4}(v_t, -L) = v_t - \frac{(3L + t)}{4} \quad (21)$$

Using eq. 16 and 21, the conditions in eq. 14, 15, 19 and 20 respectively rewrite as

$$2\sigma - \frac{3L}{4} > \frac{t}{4} + u \quad (22)$$

$$\sigma - \frac{3L}{4} \geq \frac{t}{4} \quad (23)$$

$$2\sigma - \frac{3L}{4} > \frac{u}{4} + t \quad (24)$$

$$\sigma - \frac{3L}{4} \geq \frac{u}{4} \quad (25)$$

If $\sigma < \frac{3L}{4}$, then eq. 23 (resp. 25) and 18 (resp. 13) are in contradiction. Hence, traders never adopt weakly aggressive strategies when $\sigma < \frac{3L}{4}$. If $\sigma \geq \frac{3L}{4}$, the solutions to programs 12-15 and 17-20 are given by $u_{Cases2\&5} = t_{Cases2\&5} = 0$. Hence, if

$$\sigma \geq \frac{3L}{4}$$

and traders place limit orders and follow a weakly aggressive quotation strategy, their optimal bid and offer prices are given by

$$B_{Case2\&5}(v_t, +L) = v_t - \sigma \quad (26)$$

and

$$A_{Case2\&5}(v_t, -L) = v_t + \sigma \quad (27)$$

Traders' expected profit is the same for buyers and sellers and writes

$$U_l(Cases2\&5, +L) = U_l(Cases2\&5, -L) = \frac{L}{4}$$

Traders cutoff prices are given by

$$C_{Case2\&5}(v_t, +L) = v_t + \frac{3L}{4} \quad (28)$$

$$C_{Case2\&5}(v_t, -L) = v_t - \frac{3L}{4} \quad (29)$$

3. Characterization of traders' optimal quotation strategies when buyers follow an aggressive strategy and sellers follow a weakly aggressive strategy (Cases 1 and 5)

Since the proof proceeds in a similar way, for brevity, we only report the traders' optimal quotation strategies and their expected profits. First, in cases 1 and 5, there does not exist profitable limit order strategies when $\sigma < \frac{L}{2}$ or $\sigma > \frac{7L}{4}$. Otherwise, for $\frac{L}{2} \leq \sigma \leq \frac{7L}{4}$, the traders' optimal limit prices are given by

$$\begin{aligned} B_{Case1\&5}(v_t, +L) &= v_t + \max \left\{ 0, \sigma - \frac{3L}{4} \right\} \\ A_{Case1\&5}(v_t, -L) &= v_t + \sigma \end{aligned}$$

and their expected profits write

$$\begin{aligned} U_l(Cases1\&5, +L) &= \frac{1}{2} \left[L - \max \left\{ 0, \sigma - \frac{3L}{4} \right\} \right] \\ U_l(Cases1\&5, -L) &= \frac{L}{4} \end{aligned}$$

4. Characterization of traders' optimal quotation strategies when buyers follow a weakly aggressive strategy and sellers follow an aggressive strategy (Cases 2 and 4)

Similarly, such strategies yield positive expected profits to traders iff $\frac{L}{2} \leq \sigma \leq \frac{7L}{4}$. Their optimal quotation strategies are given by

$$\begin{aligned} B_{Case2\&4}(v_t, +L) &= v_t - \sigma \\ A_{Case2\&4}(v_t, -L) &= v_t - \max \left\{ 0, \sigma - \frac{3L}{4} \right\} \end{aligned}$$

and their expected profits write

$$\begin{aligned} U_l(Cases2\&4, +L) &= \frac{L}{4} \\ U_l(Cases2\&4, -L) &= \frac{1}{2} \left[L - \max \left\{ 0, \sigma - \frac{3L}{4} \right\} \right] \end{aligned}$$

5. Traders' optimal quotation strategies

Given the traders' expected profits associated with either an aggressive or a weakly aggressive strategy, it is possible to determine whether traders optimally adopt aggressive or weakly aggressive strategies if they choose to place a limit order. We here characterize buyers' (resp. sellers') best response to sellers' (resp. buyers') quotation strategy.

First, notice that for $0 \leq \sigma < \frac{L}{2}$, the only quotation strategies which yield traders positive profits are the aggressive ones. Hence traders' optimal quotation strategies are given by eq. 8 and 9 and their execution probability is $1/2$.

For $\frac{L}{2} \leq \sigma < \frac{3L}{4}$, buyers and sellers do not both adopt weakly aggressive quotation strategies since these yield them negative profits. Moreover, observe that

$$U_l(\text{Cases1\&5}, +L) \geq U_l(\text{Cases1\&4}, +L) > U_l(\text{Cases2\&4}, +L)$$

Hence, if the quotation strategy of sellers is aggressive (Case 4), the best response of buyers is to follow an aggressive strategy (Case 1). Overall, buyers would prefer that sellers adopt a weakly aggressive strategy (Case 5) ; in this case, they would choose an aggressive strategy (Case 1). With respect to the best quotation strategy of sellers, observe that

$$U_l(\text{Cases2\&4}, -L) \geq U_l(\text{Cases1\&4}, -L) > U_l(\text{Cases1\&5}, -L)$$

Hence, if the quotation strategy of buyers is aggressive (Case 1), the best response of sellers is to follow an aggressive strategy (Case 4). Overall, sellers would prefer that buyers adopt a weakly aggressive strategy (Case 2) ; in this case, they choose an aggressive strategy (Case 4). It follows from these comparisons that the only Nash equilibrium is such that buyers and sellers both adopt aggressive quotation strategies (Cases 1 and 4). These are given by eq. 8 and 9 and their execution probability is $1/2$.

For $\frac{3L}{4} \leq \sigma < \frac{5L}{4}$, let us observe that buyers' and sellers' expected profits verify

$$\begin{aligned} U_l(\text{Cases1\&5}, +L) &> U_l(\text{Cases1\&4}, +L) > \max \{U_l(\text{Cases2\&4}, +L), U_l(\text{Cases2\&5}, +L)\} \\ U_l(\text{Cases2\&4}, -L) &> U_l(\text{Cases1\&4}, -L) > \max \{U_l(\text{Cases1\&5}, -L), U_l(\text{Cases2\&5}, -L)\} \end{aligned}$$

Hence, the only Nash equilibrium is such that buyers and sellers both adopt aggressive quotation strategies (Cases 1 and 4).

For $\frac{5L}{4} \leq \sigma < \frac{7L}{4}$, notice that

$$\begin{aligned} U_l(\text{Cases2\&5}, +L) &= U_l(\text{Cases2\&4}, +L) \geq U_l(\text{Cases1\&4}, +L) \geq U_l(\text{Cases1\&5}, +L) \\ U_l(\text{Cases2\&5}, -L) &= U_l(\text{Cases1\&5}, -L) \geq U_l(\text{Cases1\&4}, -L) \geq U_l(\text{Cases2\&4}, -L) \end{aligned}$$

Thus, the only Nash equilibrium is such that buyers and sellers both adopt weakly aggressive quotation strategies (Cases 2 and 5). These are given by eq. 26 and 27 and their execution probability is $1/4$.

For $\frac{7L}{4} \leq \sigma < 2L$, buyers and sellers do not adopt respectively aggressive (resp. weakly aggressive) and weakly aggressive (resp aggressive) quotation strategies since these would yield them negative profits. Besides, the following inequalities

$$\begin{aligned} U_l(\text{Cases2\&5}, +L) &> U_l(\text{Cases1\&4}, +L) \\ U_l(\text{Cases2\&5}, -L) &> U_l(\text{Cases1\&4}, -L) \end{aligned}$$

show that buyers and sellers optimally follow weakly aggressive strategies (Cases 2 and 5).

Finally, for $2L \leq \sigma$, the only quotation strategies which yield positive profits to both buyers and sellers is the weakly aggressive strategies (Cases 2 and 5).

- *Step 3 : Characterization of traders' optimal order choice strategies*

Step 2 identifies the optimal limit prices posted by traders if they opt for a limit order. Whether they actually opt for a limit order or a market order depends on the best price available on the opposite side of the market when they arrive to the marketplace and on its position relative to their cutoff price.

1. The market parameters verify $\sigma < \frac{5L}{4}$

Suppose that $\sigma < \frac{5L}{4}$ and consider a buyer who arrives at time t . The best ask price available in the book can *a priori* be posted either by the trader who arrived at time $(t-1)$ or by the market maker at time $(t-1)^+$.

Let us assume that at time $(t-1)$ a seller arrives and submits a sell limit order. Step 2 shows that the seller adopts an aggressive quotation strategy and sets an ask price equal to $A_{\text{Case1\&4}}(v_{t-1}, -L) = v_{t-1} - \max\left\{0, \frac{(2\sigma-L)}{3}\right\}$. Using eq. 10, it can be verified that

$A_{Case1\&4}(v_{t-1}, -L) \leq C_{Case1\&4}(v_t, +L)$. It follows that the buyer optimally submits a buy market order at time t , whether the asset value increases ($v_t = v_{t-1} + \sigma$) or decreases ($v_t = v_{t-1} - \sigma$).

If the best ask price in the book at time t has instead been set by a market maker at time $(t-1)^+$, it must be that the sell side of the book only contains the market maker's order (this is the case if the trader at time $(t-1)$ is a buyer or a seller who submits a market order). Indeed, if a seller opts for a limit order, step 2 shows that his offer is such that the next market maker cannot profitably undercut. The market maker's ask price is then determined as in Step 1-Case 6. Using eq. 10, it can be verified that $C_{Case1\&4}(v_{t-1} - \sigma, +L) > \frac{1}{2} [C_{Case1\&4}(v_{t-1} + \sigma, +L) + v_{t-1} - \sigma]$ iff $\sigma < \frac{L}{4}$. It can also be checked that $C_{Case1\&4}(v_{t-1} - \sigma, +L) > v_{t-1}$ if $\sigma < \frac{L}{4}$ and that $C_{Case1\&4}(v_{t-1} + \sigma, +L) > v_t + \sigma$ if $\frac{L}{4} \leq \sigma < \frac{5L}{4}$. Hence at time $(t-1)^+$ the market maker sets an optimal ask price equal to $A^M(v_{t-1}) = C_{Case1\&4}(v_{t-1} - \sigma, +L)$ if $\sigma < \frac{L}{4}$ and $A^M(v_{t-1}) = C_{Case1\&4}(v_{t-1} + \sigma, +L)$ if $\frac{L}{4} \leq \sigma < \frac{5L}{4}$. It follows that for $\sigma < \frac{L}{4}$, the buyer optimally submits a market order at time t . If $\frac{L}{4} \leq \sigma < \frac{5L}{4}$, the buyer optimally places a market order in case of an increase of the asset value ($v_t = v_{t-1} + \sigma$) and a limit order in case of a decrease ($v_t = v_{t-1} - \sigma$). In the latter case, his optimal bid price is given by eq. 8.

A symmetrical proof can be constructed when a seller arrives to the market at time t . If the best bid price in the book has been posted by the trader who arrived at time $(t-1)$, then the seller optimally market sells. If the best bid has been set a market maker at time $(t-1)^+$, two cases are possible. If $\sigma < \frac{L}{4}$, the market maker's bid price is given by $B^M(v_{t-1}) = C_{Case1\&4}(v_{t-1} + \sigma, -L)$ and the seller market sells. If $\frac{L}{4} \leq \sigma < \frac{5L}{4}$, the market maker's bid price is given by $B^M(v_{t-1}) = C_{Case1\&4}(v_{t-1} - \sigma, -L)$. The seller submits a market order in case of a decrease of the asset value and a limit order given by eq. 9 in case of an increase.

It follows that if $\sigma < \frac{L}{4}$, traders only submit market orders. Market makers' bid and ask prices are given by $A^M(v_t) = C_{Case1\&4}(v_t - \sigma, +L)$ and $B^M(v_t) = C_{Case1\&4}(v_t + \sigma, -L)$. The execution probability of these offers is $1/2$. If $\frac{L}{4} \leq \sigma < \frac{5L}{4}$, best prices in the book can be set by traders or market makers. When traders opt for a limit order, their quotation strategy is aggressive (cases 1 and 4) and their execution probability is $1/2$. When market makers set the best prices in the book, these are given by $A^M(v_t) = C_{Case1\&4}(v_t + \sigma, +L)$ and $B^M(v_t) = C_{Case1\&4}(v_t - \sigma, -L)$ and their execution probability is $1/4$.

2. The market parameters verify $\frac{5L}{4} \leq \sigma$

Consider a seller who arrives at time t . The best bid price available in the book can be posted

either by the trader who arrived at time $(t-1)$ or by the market maker at time $(t-1)^+$. If at time $(t-1)$ a buyer arrives and submits a limit order, Step 2 shows that the buyer adopts a weakly aggressive quotation strategy and sets a bid price equal to $B_{Case2\&4}(v_{t-1}, +L) = v_{t-1} - \sigma$. Using eq. 29, we obtain that $B_{Case2\&4}(v_{t-1}, +L) \geq C_{Case2\&5}(v_t, -L)$ if $v_t = v_{t-1} - \sigma$ and $B_{Case2\&4}(v_{t-1}, +L) < C_{Case2\&5}(v_t, -L)$ if $v_t = v_{t-1} + \sigma$. Hence, at time t the seller submits a market order in case of a decrease of the asset value and a limit order in case of an increase. In the latter case, the seller's optimal ask price is given by eq. 27 and has an execution probability equal to $1/4$.

If the trader at time $(t-1)$ is a seller or a buyer who submits a market order, then the only bid price in the book at time t has been set by a market maker at time $(t-1)^+$. Using Step 1-Case 3 and eq. 29, we obtain that the market maker's optimal bid price is $B^M(v_{t-1}) = C_{Cases2\&5}(v_{t-1} - \sigma, -L)$. It follows that the seller optimally submits a market order in case of a decrease of the asset value and a limit order in case of an increase.

A symmetrical proof can be constructed when a buyer arrives to the market at time t . Whether the best ask price posted in the book just before t has been set by a trader or a market maker, the buyer submits a market order in case of an increase of the asset value and a limit order in case of a decrease. In the latter case, his optimal bid price is given by eq. 26 and his order's execution probability is $1/4$. Using Step 1- Case 6, it can be verified that when the best ask price is set by a market maker, it is given by $A^M(v_{t-1}) = C_{Cases2\&5}(v_{t-1} + \sigma, +L)$. The preceding reasoning shows that market makers' orders have an execution probability equal to $1/4$.

For all values of the parameters (L, σ) , the optimal strategies of agents do not depend on the time at which they are submitted. Hence the unique equilibrium of the limit order market with designated market makers is stationary.

Proof of Proposition 2

It proceeds in a similar way as the proof of Proposition 3 in Foucault (1999) with $\rho = 1$. The proof first determines the optimal quotation and order choice strategies of a buyer (seller) when the sellers' (buyers') order choice strategy is taken as given. Then, buyers' and sellers' optimal responses are used to compute the optimal limit prices posted in the book at stationary equilibria. Optimal strategies are derived under the conjecture that cutoff prices $C^p(v_t, +L)$ and $C^p(v_t, -L)$ are increasing in the asset value v_t . Then it is verified that this conjecture is

indeed correct.

- *Step 1 : Characterization of buyers' optimal strategies given the order choice strategy of sellers*

Let $C^p(., -L)$ be the sellers' equilibrium cutoff value. Consider a buyer who arrives to the market at time t and places a buy limit order at a price B . The execution probability of this order depends on the position of the bid price B relative to the sell cutoff value of the trader who arrives at time $(t + 1)$. Two cases are possible. The bid price entered by the trader can either be such that $B \geq C^p(v_t + \sigma, -L) > C^p(v_t - \sigma, -L)$ (Case 1) or $C^p(v_t + \sigma, -L) > B \geq C^p(v_t - \sigma, -L)$ (Case 2). Any other bid price such that $C^p(v_t - \sigma, -L) > B$ has indeed a zero execution probability and is then never optimal.

In Case 1, any bid price higher than $C^p(v_t + \sigma, -L)$ has the same execution probability, which is the probability that a seller arrives at time $(t + 1)$. Indeed, as soon as a seller faces a bid price higher than $C^p(v_t + \sigma, -L)$, he submits a market order whether the asset value increases ($v_{t+1} = v_t + \sigma$) or decreases ($v_{t+1} = v_t - \sigma$). Hence, the buyer at time t optimally posts a bid price equal to $C^p(v_t + \sigma, -L)$ and gets the expected utility $U^p(\text{Case1}, +L) = \frac{1}{2} [v_t + L - C^p(v_t + \sigma, -L)]$. This strategy is profitable iff $C^p(v_t + \sigma, -L) < (v_t + L)$.

In Case 2, any bid price in the interval $[C^p(v_t - \sigma, -L), C^p(v_t + \sigma, -L)[$ has the same execution probability. It is executed only if the asset value decreases ($v_{t+1} = v_t - \sigma$) and a seller arrives at time $(t + 1)$. Hence, the buyer at time t optimally posts the smallest bid price in this interval, that is a price equal to $C^p(v_t - \sigma, -L)$. He gets the expected utility $U^p(\text{Case2}, +L) = \frac{1}{4} [v_t - \sigma + L - C^p(v_t - \sigma, -L)]$. Such a strategy is profitable iff $C^p(v_t - \sigma, -L) < (v_t - \sigma + L)$.

If both strategies played in Cases 1 and 2 yield positive expected profits to the buyer, the latter chooses to place the bid price $C^p(v_t + \sigma, -L)$ iff $U^p(\text{Case1}, +L) > U^p(\text{Case2}, +L)$ and chooses $C^p(v_t - \sigma, -L)$ otherwise. After rewriting the last inequality, the buyer chooses a bid price equal to $C^p(v_t + \sigma, -L)$ iff

$$C^p(v_t + \sigma, -L) < \frac{1}{2} [C^p(v_t - \sigma, -L) + v_t + \sigma + L] \quad (30)$$

and $C^p(v_t - \sigma, -L)$ otherwise.

Given the definition of the buyer's cutoff value in eq. 1 and the expected profits earned by

the trader, his cutoff values are respectively given in Cases 1 and 2 by :

$$C_{Case1}^p(v_t, +L) = \frac{1}{2} [v_t + L + C^p(v_t + \sigma, -L)] \quad (31)$$

$$C_{Case2}^p(v_t, +L) = \frac{1}{4} [3(v_t + L) + \sigma + C^p(v_t - \sigma, -L)] \quad (32)$$

- *Step 2 : Characterization of sellers' optimal strategies given the order choice strategy of buyers*

Consider a seller who arrives to the market at time $(t + 1)$ and places a sell limit order at a price A . Let $C^p(., +L)$ be the buyers' equilibrium cutoff value. Proceeding as in Step 1, two different quotation strategies may result optimal for the seller. The ask price posted by the trader may either be such that $A \leq C^p(v_{t+1} - \sigma, +L) < C^p(v_{t+1} + \sigma, +L)$ (Case 3) or $C^p(v_{t+1} - \sigma, +L) < A \leq C^p(v_{t+1} + \sigma, +L)$ (Case 4). In Case 3, since any sell price lower than $C^p(v_{t+1} - \sigma, +L)$ has an execution probability of $1/2$, the trader optimally submits a sell order at a price equal to $C^p(v_{t+1} - \sigma, +L)$ and obtain the expected profit $U^p(Case3, -L) = \frac{1}{2} [C^p(v_{t+1} - \sigma, +L) - (v_{t+1} - L)]$. In Case 4, any ask price in the interval $[C^p(v_{t+1} - \sigma, +L), C^p(v_{t+1} + \sigma, +L)]$ has the same execution probability of $1/4$. It is executed only if the asset value increases ($v_{t+2} = v_{t+1} + \sigma$) and a buyer arrives at time $(t + 2)$. Hence, the seller at time $(t + 1)$ optimally places an ask price equal to $C^p(v_{t+1} + \sigma, +L)$. He gets the expected profit $U^p(Case4, -L) = \frac{1}{4} [C^p(v_{t+1} + \sigma, +L) - (v_{t+1} + \sigma - L)]$.

If both strategies played in Cases 3 and 4 yield positive expected profits to the seller, the latter chooses to place an ask price equal to $C^p(v_{t+1} - \sigma, +L)$ iff $U^p(Case3, -L) > U^p(Case4, -L)$ and chooses $C^p(v_{t+1} + \sigma, +L)$ otherwise. After rewriting the last inequality, the seller at time $(t + 1)$ chooses an ask price equal to $C^p(v_{t+1} - \sigma, +L)$ iff

$$C^p(v_{t+1} - \sigma, +L) > \frac{1}{2} [C^p(v_{t+1} + \sigma, +L) + v_{t+1} - \sigma - L] \quad (33)$$

and $C^p(v_{t+1} + \sigma, +L)$ otherwise.

Given the definition of the seller's cutoff value in eq. 2 and the expected profits earned by the trader, his cutoff values are respectively given in Cases 3 and 4 by :

$$C_{Case3}^p(v_{t+1}, -L) = \frac{1}{2} [v_{t+1} - L + C^p(v_{t+1} - \sigma, +L)] \quad (34)$$

$$C_{Case4}^p(v_{t+1}, -L) = \frac{1}{4} [3(v_{t+1} - L) - \sigma + C^p(v_{t+1} + \sigma, +L)] \quad (35)$$

- *Step 3 : Traders' optimal strategies at stationary equilibria*

Step 3.1 : The market parameters verify $\sigma < \frac{4L}{5}$

Suppose that the conditions in eq. 30 and 33 are satisfied (Cases 1 and 3). It follows from Steps 1 and 2 that the optimal bid (ask) price posted at time t ($(t+1)$) by a buyer (seller) who opts for a limit order is equal to $C_{Case3}^p(v_t + \sigma, -L)$ ($C_{Case1}^p(v_{t+1} - \sigma, +L)$). Using eq. 34 and 31, these cutoff values verify

$$\begin{aligned} C_{Case3}^p(v_t + \sigma, -L) &= \frac{1}{2} [v_t + \sigma - L + C_{Case1}^p(v_t, +L)] \\ C_{Case1}^p(v_{t+1} - \sigma, +L) &= \frac{1}{2} [v_{t+1} - \sigma + L + C_{Case3}^p(v_{t+1}, -L)] \end{aligned} \quad (36)$$

In case of an increase of the asset value ($v_{t+1} = v_t + \sigma$), the last equation rewrites as :

$$C_{Case1}^p(v_t, +L) = \frac{1}{2} [v_t + L + C_{Case3}^p(v_t + \sigma, -L)] \quad (37)$$

The combination of eq. 36 and 37 yields

$$C_{Case1}^p(v_t, +L) = v_t + \frac{(L + \sigma)}{3} \quad (38)$$

Proceeding in the same way

$$C_{Case3}^p(v_t, -L) = v_t - \frac{(L + \sigma)}{3} \quad (39)$$

Using the expressions of the cutoff values given by eq. 38 and 39, we obtain that the conditions in eq. 30 and 33 are verified iff $\sigma < \frac{4L}{5}$. Consequently, for these market parameters, an equilibrium quotation strategy of traders who choose to place a limit order is given by

$$\begin{aligned} A^p(v_t, -L) &= C_{Case1}^p(v_t - \sigma, +L) = v_t - \frac{(2\sigma - L)}{3} \\ B^p(v_t, +L) &= C_{Case3}^p(v_t + \sigma, -L) = v_t + \frac{(2\sigma - L)}{3} \end{aligned}$$

and the execution probability of these offers is $1/2$.

Step 3.2 : The market parameters verify $\sigma \geq \frac{4L}{5}$

Suppose that the conditions in eq. 30 and 33 are not verified (Cases 2 and 4). Hence,

$$C^p(v_t + \sigma, -L) \geq \frac{1}{2} [C^p(v_t - \sigma, -L) + v_t + \sigma + L] \quad (40)$$

$$C^p(v_{t+1} - \sigma, +L) \leq \frac{1}{2} [C^p(v_{t+1} + \sigma, +L) + v_{t+1} - \sigma - L] \quad (41)$$

It follows from Steps 1 and 2 that the optimal bid (ask) price posted at time t ($(t+1)$) by a buyer (seller) who opts for a limit order is equal to $C_{Case4}^p(v_t - \sigma, -L)$ ($C_{Case2}^p(v_{t+1} + \sigma, +L)$). Using eq. 35 and 32, these cutoff values verify

$$\begin{aligned} C_{Case4}^p(v_t - \sigma, -L) &= \frac{1}{4} [3(v_t - L) - 4\sigma + C_{Case2}^p(v_t, +L)] \\ C_{Case2}^p(v_{t+1} + \sigma, +L) &= \frac{1}{4} [3(v_{t+1} + L) + 4\sigma + C_{Case4}^p(v_{t+1}, -L)] \end{aligned}$$

This system can be used to solve for the buy and sell cutoff values. Proceeding as in Step 3.1, we obtain

$$C_{Case2}^p(v_t, +L) = v_t + \frac{3}{5}L \quad (42)$$

$$C_{Case4}^p(v_t, -L) = v_t - \frac{3}{5}L \quad (43)$$

Using these expressions of the cutoff values, it turns out that the conditions in eq. 40 and 41 are satisfied iff $\sigma \geq \frac{4L}{5}$. Consequently, for these market parameters, an equilibrium quotation strategy of traders who choose to place a limit order is given by

$$\begin{aligned} A^p(v_t, -L) &= C_{Case2}^p(v_t + \sigma, +L) = v_t + \sigma + \frac{3}{5}L \\ B^p(v_t, +L) &= C_{Case4}^p(v_t - \sigma, -L) = v_t - \sigma - \frac{3}{5}L \end{aligned}$$

The execution probability of these offers is $1/4$.

It is possible to show that no other equilibrium than the one derived above, can be obtained. Cases 1 to 4 enable to identify two regimes for the traders' quotation strategies : in Cases 1 and 3, buyers' and seller's quotation strategies are *aggressive* while they are *weakly aggressive* in Cases 2 and 4. Due to symmetrical assumptions across buyers and sellers, it is never optimal for a buyer to adopt an aggressive (a weakly aggressive) quotation strategy when sellers choose a weakly aggressive (an aggressive) one. Similarly, the optimal quotation strategy of a seller when buyers' quotes are aggressive (weakly aggressive) does not involve the submission of weakly aggressive (aggressive) prices. Hence, the only equilibrium is such that buyers and sellers both adopt aggressive (Cases 1 and 3) or weakly aggressive (Cases 2 and 4) quotation strategies. The computations necessary to show the uniqueness of the equilibrium are omitted for brevity.

Proof of Proposition 3

Let $p_i(MO)$ ($p_i(LO)$) be the probability that the trader who arrives at time i submits a

market (limit) order. Then $m_t = \frac{1}{(t-1)} \sum_{i=0}^{(t-2)} p_i(MO)$ and

$$\begin{aligned} p_i(MO) &= \Pr(I_i(MO) = 1 \mid I_{i-1}(MO) = 1) p_{i-1}(MO) \\ &\quad + \Pr(I_i(MO) = 1 \mid I_{i-1}(LO) = 1) p_{i-1}(LO) \end{aligned} \quad (44)$$

Let also $I_i(BuyMO)$ and $I_i(SellMO)$ (resp. $I_i(BuyLO)$ and $I_i(SellLO)$) be indicator functions which respectively take the value +1 if the trader who arrives at time i submits a buy and a sell market order (resp. a buy and a sell limit order) and 0 otherwise. Since a trader can be either a buyer or a seller, we can write

$$\begin{aligned} \Pr(I_i(MO) = 1 \mid I_{i-1}(MO) = 1) &= \Pr(I_i(BuyMO) = 1 \mid I_{i-1}(MO) = 1) \\ &\quad + \Pr(I_i(SellMO) = 1 \mid I_{i-1}(MO) = 1) \end{aligned} \quad (45)$$

and

$$\begin{aligned} \Pr(I_i(MO) = 1 \mid I_{i-1}(LO) = 1) &= \Pr(I_i(BuyMO) = 1 \mid I_{i-1}(LO) = 1) \\ &\quad + \Pr(I_i(SellMO) = 1 \mid I_{i-1}(LO) = 1) \end{aligned} \quad (46)$$

If $0 \leq \sigma < \frac{L}{4}$, traders only place limit orders. Hence, $p_i(MO) = 1$ and $m = m_t = 1$.

If $\frac{L}{4} \leq \sigma$ and if at time $(i-1)$ the trader places a market order ($I_{i-1}(MO) = 1$), the best offers in the book at time i are set by the market maker who arrives at time $(i-1)^+$. $\Pr(I_i(BuyMO) = 1 \mid I_{i-1}(MO) = 1)$ is thus the execution probability of the market maker's sell limit order at time $(i-1)^+$. From Proposition 1, we know that this execution probability is $1/4$ if $\sigma \geq \frac{L}{4}$. Since the buy and sell sides are completely symmetric in this model, $\Pr(I_i(SellMO) = 1 \mid I_{i-1}(MO) = 1)$ is also equal to $1/4$. Hence, using eq. 45

$$\Pr(I_i(MO) = 1 \mid I_{i-1}(MO) = 1) = \frac{1}{2}$$

Let us compute $\Pr(I_i(BuyMO) = 1 \mid I_{i-1}(LO) = 1)$. If at time $(i-1)$ the trader places a limit order ($I_{i-1}(LO) = 1$), two cases are possible. First, this trader is a buyer and places a buy limit order. This happens with probability $1/2$. At time i , the best ask price is then set by a market maker and $\Pr(I_i(BuyMO) = 1 \mid I_{i-1}(BuyLO) = 1)$ is the execution probability of the market maker's sell limit order at time $(i-1)^+$. From Proposition 1, this execution probability is $1/4$ if $\sigma \geq \frac{L}{4}$. Second, the trader at time $(i-1)$ is a seller. At time i , the best ask price is then set by this trader and $\Pr(I_i(BuyMO) = 1 \mid I_{i-1}(SellLO) = 1)$ is the execution probability of

a trader's sell limit order at time $(i - 1)$. From Proposition 1, we know that this execution probability is $1/2$ if $\frac{L}{4} \leq \sigma < \frac{5L}{4}$ and $1/4$ if $\sigma \geq \frac{5L}{4}$. Hence if $\frac{L}{4} \leq \sigma$

$$\begin{aligned} \Pr(I_i(\text{BuyMO}) = 1 \mid I_{i-1}(LO) = 1) &= \frac{1}{2} \Pr(I_i(\text{BuyMO}) = 1 \mid I_{i-1}(\text{BuyLO}) = 1) \\ &\quad + \frac{1}{2} \Pr(I_i(\text{BuyMO}) = 1 \mid I_{i-1}(\text{SellLO}) = 1) \quad (47) \\ &= \frac{1}{2} * \frac{1}{4} + \frac{1}{2} \Pr(I_i(\text{BuyMO}) = 1 \mid I_{i-1}(\text{SellLO}) = 1) \end{aligned}$$

If $\frac{L}{4} \leq \sigma < \frac{5L}{4}$,

$$\Pr(I_i(\text{BuyMO}) = 1 \mid I_{i-1}(LO) = 1) = \frac{1}{8} + \frac{1}{2} * \frac{1}{2} = \frac{3}{8}$$

while if $\sigma \geq \frac{5L}{4}$

$$\Pr(I_i(\text{BuyMO}) = 1 \mid I_{i-1}(LO) = 1) = \frac{1}{8} + \frac{1}{2} * \frac{1}{4} = \frac{1}{4}$$

Since the buy and sell sides are symmetric, $\Pr(I_i(\text{SellMO}) = 1 \mid I_{i-1}(LO) = 1)$ is equal to $\Pr(I_i(\text{BuyMO}) = 1 \mid I_{i-1}(LO) = 1)$. Hence, using eq. 46

$$\begin{aligned} \Pr(I_i(MO) = 1 \mid I_{i-1}(LO) = 1) &= \frac{3}{4} && \text{if } \frac{L}{4} \leq \sigma < \frac{5L}{4} \\ \Pr(I_i(MO) = 1 \mid I_{i-1}(LO) = 1) &= \frac{1}{2} && \text{if } \frac{5L}{4} \leq \sigma \end{aligned}$$

Finally, using eq. 44 and the fact that $p_{i-1}(LO) = 1 - p_{i-1}(MO)$, we can write for $\frac{L}{4} \leq \sigma < \frac{5L}{4}$

$$p_i(MO) = \frac{1}{2} p_{i-1}(MO) + \frac{3}{4} [1 - p_{i-1}(MO)]$$

This yields

$$m_t = \frac{3}{4} - \frac{(t-2)}{4(t-1)} m_{t-1}$$

Taking the limit on both sides when t goes to infinity, we obtain $m = 3/5$. Using the same reasoning $\sigma \geq \frac{5L}{4}$, we get in this case $m = 1/2$.

Proof of Proposition 4

It proceeds as the proof of Proposition 4 in Foucault (1999). Let m_t^p and m^p respectively denote the proportion of market orders in the order flow in the interval of time $[0, (t - 2)]$ and at the stationary equilibrium in a pure limit order market. Apart from these, we use the same notations than in the proof of Proposition 3. In the limit order market without designated market makers, if the trader who arrives at time $(i - 1)$ submits a market order, then the order book at time i is empty. Hence, the trader who arrives at time i cannot place a market order. It follows that

$$\Pr(I_i(MO) = 1 \mid I_{i-1}(MO) = 1) = 0$$

Let us compute $\Pr(I_i(BuyMO) = 1 \mid I_{i-1}(LO) = 1)$. If at time $(i-1)$ the trader places a limit order ($I_{i-1}(LO) = 1$), two cases are possible. First, with probability $1/2$, this trader is a seller. At time i , the sell side of the order book is then non empty and $\Pr(I_i(BuyMO) = 1 \mid I_{i-1}(SellLO) = 1)$ is the execution probability of a sell limit order at time $(i-1)$. From Proposition 2, we know that this execution probability is equal to $1/2$ if $0 \leq \sigma < \frac{4L}{5}$ and $1/4$ if $\sigma \geq \frac{4L}{5}$. Second, with probability $1/2$, the trader at time $(i-1)$ is a buyer. At time i , the sell side of the order book is then empty. Hence, the trader who arrives at time i cannot place a buy market order. It follows that $\Pr(I_i(BuyMO) = 1 \mid I_{i-1}(BuyLO) = 1) = 0$. Proceeding as in eq. 47, we obtain

$$\begin{aligned} \Pr(I_i(BuyMO) = 1 \mid I_{i-1}(LO) = 1) &= \frac{1}{4} && \text{if } 0 \leq \sigma < \frac{4L}{5} \\ \Pr(I_i(BuyMO) = 1 \mid I_{i-1}(LO) = 1) &= \frac{1}{8} && \text{if } \frac{4L}{5} \leq \sigma \end{aligned}$$

Since the buy and sell sides are symmetric,

$$\begin{aligned} \Pr(I_i(MO) = 1 \mid I_{i-1}(LO) = 1) &= \frac{1}{2} && \text{if } 0 \leq \sigma < \frac{4L}{5} \\ \Pr(I_i(MO) = 1 \mid I_{i-1}(LO) = 1) &= \frac{1}{4} && \text{if } \frac{4L}{5} \leq \sigma \end{aligned}$$

Then, using eq. 44, the definition of m_t^p and the fact that $p_{i-1}(LO) = 1 - p_{i-1}(MO)$, we can write

$$\begin{aligned} m_t^p &= \frac{1}{2} - \frac{(t-2)}{2(t-1)} m_{t-1}^p && \text{if } 0 \leq \sigma < \frac{4L}{5} \\ m_t^p &= \frac{1}{4} - \frac{(t-2)}{4(t-1)} m_{t-1}^p && \text{if } \frac{4L}{5} \leq \sigma \end{aligned}$$

Taking the limit when t goes to infinity on both sides of these two equations yields $m^p = 1/3$ if $0 \leq \sigma < \frac{4L}{5}$ and $m^p = 1/5$ if $\frac{4L}{5} \leq \sigma$.

Proof of Corollary 1

It results directly from Propositions 3 and 4. The following table reports the equilibrium proportions of market orders in the order flow in the cases of a limit order market with market makers (m) and without them (m^p).

	$0 \leq \sigma < \frac{L}{4}$	$\frac{L}{4} \leq \sigma < \frac{4L}{5}$	$\frac{4L}{5} \leq \sigma < \frac{5L}{4}$	$\frac{5L}{4} \leq \sigma$
m	1	3/5		1/2
m^p	1/3		1/5	

It can be verified that for all values of the parameters (L, σ) , m is strictly larger than m^p .

Proof of Proposition 5

Proposition 5 follows from Propositions 1 and 2. The following table reports the best ask prices set by agents for all values of the parameters (L, σ) . It can be verified that $A(v_t, -L) < A^p(v_t, -L)$ if $\frac{L}{4} \leq \sigma < \frac{L}{2}$ and $\sigma \geq \frac{4L}{5}$ while $A(v_t, -L) = A^p(v_t, -L)$ if $\frac{L}{2} \leq \sigma < \frac{4L}{5}$. Furthermore, $A^M(v_t) > A^p(v_t, -L)$ for all values of the parameters except for $\sigma = \frac{4L}{5}$ and $A^M(v_t) = A^p(v_t, -L)$ if $\sigma = \frac{4L}{5}$. Since bid and ask prices are symmetric around v_t for all agents, these results also yield that $B(v_t, +L) \geq B^p(v_t, +L)$ and $B^M(v_t) \leq B^p(v_t, +L)$ for all values of the parameters.

	$0 \leq \sigma < \frac{L}{4}$	$\frac{L}{4} \leq \sigma < \frac{4L}{5}$	$\frac{4L}{5} \leq \sigma < \frac{5L}{4}$	$\frac{5L}{4} \leq \sigma$
$A^M(v_t)$	$v_t - \sigma + \frac{L}{2}$	$v_t + \sigma + \frac{L}{2} + \max\left\{0, \frac{(2\sigma-L)}{6}\right\}$		$v_t + \sigma + \frac{3L}{4}$
$A(v_t, -L)$		$v_t - \max\left\{0, \frac{(2\sigma-L)}{3}\right\}$		$v_t + \sigma$
$A^p(v_t, -L)$	$v_t - \frac{(2\sigma-L)}{3}$		$v_t + \sigma + \frac{3}{5}L$	

Proof of Proposition 6

Since a symmetrical proof can be constructed for ask prices, we only focus on the bid side of the book. We know from the proof of Proposition 3 that the stationary probability that a trader chooses a buy limit order is the proportion of buy limit orders in the order flow. Let $p(\text{BuyLO})$ denote the stationary probability that a trader chooses a buy limit order. Using the symmetry between the behaviour of sellers and buyers yields

$$p(\text{BuyLO}) = \frac{(1-m)}{2}$$

where $(1-m)$ is by definition the proportion of limit orders in the order flow. Just before time t , the expected best bid price defined in eq. 3 is then given by

$$B(v_{t-1}) = \frac{(1-m)}{2} B(v_{t-1}, +L) + \left[1 - \frac{(1-m)}{2}\right] B^M(v_{t-1}) \quad (48)$$

Propositions 1 and 3 then enable to compute $B(v_{t-1})$ for all values of the parameters.

If $0 \leq \sigma < \frac{L}{4}$, traders always submit market orders in the limit order market with market makers : $m = 1$. Hence, $B(v_{t-1})$ is equal to $B^M(v_{t-1})$ and the proof of Proposition 5 shows that $B^M(v_{t-1})$ is strictly lower than $B^p(v_{t-1}, +L)$.

If $\frac{L}{4} \leq \sigma < \frac{4L}{5}$, the proportion of limit orders in the order flow is equal to $2/5$ in the limit order market with market makers. If $\frac{L}{4} \leq \sigma < \frac{L}{2}$, using Proposition 1, the expected best bid price defined in eq. 48 is given by $B(v_{t-1}) = \frac{1}{5}v_{t-1} + \frac{4}{5}\left(v_{t-1} - \sigma - \frac{L}{2}\right) = v_{t-1} - \frac{4}{5}\left(\sigma + \frac{L}{2}\right)$. It

follows from Proposition 2 that $B(v_{t-1})$ is strictly lower than $B^p(v_{t-1}, +L)$. If $\frac{L}{2} \leq \sigma < \frac{4L}{5}$, Proposition 1 and 2 show that $B(v_{t-1}, +L) = B^p(v_{t-1}, +L)$. Moreover, we know from the proof of Proposition 5 that $B^M(v_{t-1}) < B^p(v_{t-1}, +L)$. Hence, $B(v_{t-1}) < B^p(v_{t-1}, +L)$.

If $\frac{4L}{5} \leq \sigma < \frac{5L}{4}$, use of Propositions 1 and 3 yields that $B(v_{t-1}) = v_{t-1} + \frac{14}{15}\sigma + \frac{1}{3}L$. It follows from Proposition 2 that $B(v_{t-1})$ is strictly higher than $B^p(v_{t-1}, +L)$. Finally, if $\frac{5L}{4} \leq \sigma$, we obtain from Propositions 1, 3 and 5 that $B(v_{t-1}) = v_{t-1} - \sigma - \frac{9}{16}L$. It follows from Proposition 2 that $B(v_{t-1}) > B^p(v_{t-1}, +L)$.

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